#1 · Selection of Nuclear Cardiology Imaging Tests

What are the three features of typical angina?

Answer

Substernal discomfort, provoked by exertion or emotional stress, and relieved by rest or nitroglycerin within minutes. Two features is atypical angina; one or none is nonanginal chest pain.

#2 · Selection of Nuclear Cardiology Imaging Tests

How is pretest probability of obstructive CAD estimated?

Answer

From age, sex and the chest pain class (typical, atypical, nonanginal), then adjusted for risk factors such as diabetes and kidney disease. Tables derived from Diamond and Forrester are the classic basis.

#3 · Selection of Nuclear Cardiology Imaging Tests

What pretest probability range is called intermediate, and why does it matter?

Answer

Roughly 15 to 85 percent. Testing changes the post-test probability most in this range, so it is where stress imaging is most useful.

#4 · Selection of Nuclear Cardiology Imaging Tests

What is the formula for post-test odds in Bayes theorem?

Answer

Post-test odds = pretest odds x likelihood ratio. Convert back with probability = odds / (1 + odds).

#5 · Selection of Nuclear Cardiology Imaging Tests

How do you convert a pretest probability p to odds?

Answer

Odds = p / (1 - p). A probability of 0.25 gives odds of 0.25 / 0.75 = 0.33.

#6 · Selection of Nuclear Cardiology Imaging Tests

Define LR+ and LR-.

Answer

LR+ = sensitivity / (1 - specificity); LR- = (1 - sensitivity) / specificity. Neither changes with disease prevalence.

#7 · Selection of Nuclear Cardiology Imaging Tests

Sensitivity 88 percent and specificity 75 percent: what are LR+ and LR-?

Answer

LR+ = 0.88 / 0.25 = 3.52. LR- = 0.12 / 0.75 = 0.16.

#8 · Selection of Nuclear Cardiology Imaging Tests

With those LRs and a 50 percent pretest probability, what is the post-test probability after a positive and after a negative test?

Answer

Positive: odds 3.52, probability 78 percent.
Negative: odds 0.16, probability 14 percent.

#9 · Selection of Nuclear Cardiology Imaging Tests

Why is a positive stress MPI of limited value at 5 percent pretest probability?

Answer

With LR+ 3.52 the post-test probability is only about 16 percent, so most positives are false. Testing low-risk patients mainly produces false positives.

#10 · Selection of Nuclear Cardiology Imaging Tests

Why does a negative test add little at 90 percent pretest probability?

Answer

Residual probability stays high and the patient will likely go to angiography or treatment anyway, so the result rarely changes the plan.

#11 · Selection of Nuclear Cardiology Imaging Tests

Define sensitivity.

Answer

TP / (TP + FN): the fraction of patients with disease who have a positive test.

#12 · Selection of Nuclear Cardiology Imaging Tests

Define specificity.

Answer

TN / (TN + FP): the fraction of patients without disease who have a negative test.

#13 · Selection of Nuclear Cardiology Imaging Tests

Define positive predictive value.

Answer

TP / (TP + FP): the fraction of test-positive patients who truly have disease. It rises with prevalence.

#14 · Selection of Nuclear Cardiology Imaging Tests

Define negative predictive value.

Answer

TN / (TN + FN): the fraction of test-negative patients who truly lack disease. It falls as prevalence rises.

#15 · Selection of Nuclear Cardiology Imaging Tests

How do predictive values change when prevalence falls?

Answer

PPV falls and NPV rises, while sensitivity and specificity stay the same.

#16 · Selection of Nuclear Cardiology Imaging Tests

Calculate PPV: 1,000 patients, 20 percent prevalence, sensitivity 90 percent, specificity 80 percent.

Answer

TP = 180, FN = 20, TN = 640, FP = 160. PPV = 180 / 340 = 52.9 percent.

#17 · Selection of Nuclear Cardiology Imaging Tests

Same group (20 percent prevalence, sens 90, spec 80): what is the NPV?

Answer

NPV = 640 / (640 + 20) = 640 / 660 = 97.0 percent.

#18 · Selection of Nuclear Cardiology Imaging Tests

What is post-test referral (verification) bias?

Answer

Only patients with abnormal scans go to angiography, which inflates sensitivity and deflates specificity in the reported data.

#19 · Selection of Nuclear Cardiology Imaging Tests

What is the normalcy rate?

Answer

The fraction of patients with a low pretest likelihood of CAD (under 5 percent) who have a normal scan. It substitutes for specificity when normals are not verified by angiography.

#20 · Selection of Nuclear Cardiology Imaging Tests

What does an ROC curve plot, and what does AUC 0.5 mean?

Answer

Sensitivity against 1 minus specificity across thresholds. An AUC of 0.5 is chance; 1.0 is perfect discrimination.

#21 · Selection of Nuclear Cardiology Imaging Tests

What is spectrum bias?

Answer

Test performance looks better when studied in patients with severe disease and clear controls than in a mixed clinical population.

#22 · Selection of Nuclear Cardiology Imaging Tests

What is the Duke treadmill score formula?

Answer

Exercise time in minutes - (5 x maximal ST deviation in mm) - (4 x angina index). The angina index is 0 none, 1 angina not limiting, 2 angina limiting the test.

#23 · Selection of Nuclear Cardiology Imaging Tests

What are the Duke score risk groups?

Answer

Low risk is +5 or higher, intermediate is -10 to +4, high is -11 or lower.

#24 · Selection of Nuclear Cardiology Imaging Tests

Compute the Duke score: 9 minutes, 2 mm ST depression, non-limiting angina.

Answer

9 - 10 - 4 = -5, intermediate risk.

#25 · Selection of Nuclear Cardiology Imaging Tests

Compute the Duke score: 6 minutes, 3 mm ST depression, angina stops the test.

Answer

6 - 15 - 8 = -17, high risk.

#26 · Selection of Nuclear Cardiology Imaging Tests

Which resting ECG findings steer a patient to vasodilator rather than exercise stress?

Answer

Left bundle branch block, ventricular pacing, and preexcitation (WPW). They make ST analysis unreliable and LBBB or pacing can produce an artifactual septal defect with exercise.

#27 · Selection of Nuclear Cardiology Imaging Tests

What are the 2013 ACC appropriate use ratings and scores?

Answer

Appropriate is 7 to 9, may be appropriate is 4 to 6, rarely appropriate is 1 to 3.

#28 · Selection of Nuclear Cardiology Imaging Tests

A rarely appropriate study is ordered. What does that mean?

Answer

It is not a ban. The study may still proceed with documented reasoning, but it is not expected to improve care for that scenario.

#29 · Selection of Nuclear Cardiology Imaging Tests

Best test for an intermediate-risk patient who cannot exercise and has stable chest pain?

Answer

Vasodilator or dobutamine stress imaging, such as MPI. The exercise ECG would not be feasible.

#30 · Selection of Nuclear Cardiology Imaging Tests

A low-risk patient has an interpretable ECG and good functional capacity. What is the best approach?

Answer

An exercise ECG alone or no testing. Stress MPI is rarely appropriate.

#31 · Selection of Nuclear Cardiology Imaging Tests

How should a CCTA stenosis of about 50 percent be handled?

Answer

It is intermediate, so functional testing such as stress MPI is appropriate to decide whether it causes ischemia.

#32 · Selection of Nuclear Cardiology Imaging Tests

Does a CAC score of zero exclude CAD?

Answer

No. It lowers the chance of obstructive disease in a low-risk patient but does not exclude noncalcified plaque.

#33 · Selection of Nuclear Cardiology Imaging Tests

When is rest MPI appropriate in acute chest pain?

Answer

When ECG is nondiagnostic, the initial troponin is negative, and the scan can be done during pain or within a few hours of it. A normal scan has NPV above 95 percent for infarction.

#34 · Selection of Nuclear Cardiology Imaging Tests

What do you do with a stable patient after myocardial infarction is ruled out and who has no high-risk features?

Answer

Stress MPI is appropriate for risk stratification.

#35 · Selection of Nuclear Cardiology Imaging Tests

Is stress testing appropriate in an unstabilized acute coronary syndrome?

Answer

No. These patients need treatment, and often angiography, not stress.

#36 · Selection of Nuclear Cardiology Imaging Tests

Name absolute contraindications to stress testing.

Answer

Acute MI within 2 days, unstabilized unstable angina, uncontrolled symptomatic arrhythmia, symptomatic severe aortic stenosis, decompensated heart failure, acute pulmonary embolism, acute myocarditis or pericarditis, and acute aortic dissection.

#37 · Selection of Nuclear Cardiology Imaging Tests

What resting BP is a relative contraindication to exercise?

Answer

Systolic above 200 or diastolic above 110 mm Hg.

#38 · Selection of Nuclear Cardiology Imaging Tests

What physical exam sign set suggests severe aortic stenosis?

Answer

A harsh crescendo-decrescendo systolic murmur with a soft second heart sound. Exercise in the symptomatic patient is contraindicated.

#39 · Selection of Nuclear Cardiology Imaging Tests

First step in preoperative evaluation for noncardiac surgery?

Answer

Emergency surgery proceeds without testing. Otherwise estimate MACE risk; below 1 percent needs no testing.

#40 · Selection of Nuclear Cardiology Imaging Tests

A patient has elevated surgical risk and can climb two flights of stairs. What next?

Answer

Functional capacity of 4 METs or more means proceed to surgery without further testing.

#41 · Selection of Nuclear Cardiology Imaging Tests

When is preoperative stress testing reasonable?

Answer

Elevated surgical risk, functional capacity under 4 METs or unknown, and the result would change management.

#42 · Selection of Nuclear Cardiology Imaging Tests

New heart failure with no known CAD: what is the role of MPI?

Answer

Appropriate for ischemic evaluation, and it also gives ejection fraction, volumes and viability information.

#43 · Selection of Nuclear Cardiology Imaging Tests

Which agents assess viability when revascularization is considered?

Answer

Rest-redistribution Tl-201, rest-delayed Tc-99m sestamibi, and F-18 FDG PET.

#44 · Selection of Nuclear Cardiology Imaging Tests

Is MPI appropriate within 5 years of bypass surgery in an asymptomatic patient?

Answer

No, it is rarely appropriate. A new symptom changes that to appropriate.

#45 · Selection of Nuclear Cardiology Imaging Tests

List the steps to decide whether an ordered test is appropriate.

Answer

Is the patient stable; what is the clinical question; will the result change management; is a simpler test enough; are the stress and tracer right; was a recent comparable test done.

#46 · Selection of Nuclear Cardiology Imaging Tests

What is the standard adenosine infusion for SPECT stress?

Answer

140 mcg/kg/min for 6 minutes, with the radiotracer injected at the end of the third minute.

#47 · Selection of Nuclear Cardiology Imaging Tests

What is the regadenoson dose and how is it given?

Answer

A fixed 0.4 mg (5 mL) IV bolus over about 10 seconds, with no weight adjustment, followed by a 5 mL saline flush.

#48 · Selection of Nuclear Cardiology Imaging Tests

When is the radiotracer injected after regadenoson?

Answer

About 10 to 20 seconds after the saline flush, because hyperemia lasts only a few minutes.

#49 · Selection of Nuclear Cardiology Imaging Tests

What receptor does regadenoson target and why does that matter?

Answer

It is a selective A2A agonist, so AV block and bronchospasm (A1 and A2B effects) are less common than with adenosine.

#50 · Selection of Nuclear Cardiology Imaging Tests

Why does adenosine cause AV block and bronchospasm?

Answer

It also acts on A1 receptors in the AV node and A2B receptors in the bronchi.

#51 · Selection of Nuclear Cardiology Imaging Tests

What is the dipyridamole stress dose?

Answer

0.142 mg/kg/min over 4 minutes (0.56 mg/kg total), with tracer 3 to 5 minutes after the infusion ends.

#52 · Selection of Nuclear Cardiology Imaging Tests

How does dipyridamole produce vasodilation?

Answer

It blocks reuptake and breakdown of endogenous adenosine, raising interstitial adenosine.

#53 · Selection of Nuclear Cardiology Imaging Tests

What reverses vasodilator side effects?

Answer

Aminophylline 50 to 250 mg IV.

#54 · Selection of Nuclear Cardiology Imaging Tests

What is the dobutamine stress ramp?

Answer

Start at 5 mcg/kg/min, then 10, 20, 30 and 40 mcg/kg/min at 3 minute stages.

#55 · Selection of Nuclear Cardiology Imaging Tests

What is added if dobutamine alone does not reach target heart rate?

Answer

Atropine in 0.25 to 0.5 mg increments, up to 1 mg total.

#56 · Selection of Nuclear Cardiology Imaging Tests

When should dobutamine stress be avoided?

Answer

With uncontrolled hypertension, significant ventricular arrhythmia, aortic dissection or an unstable aneurysm.

#57 · Selection of Nuclear Cardiology Imaging Tests

What is the target heart rate for an adequate exercise or dobutamine study?

Answer

85 percent of age predicted maximum heart rate (220 minus age).

#58 · Selection of Nuclear Cardiology Imaging Tests

A patient has LBBB and can walk well. What stressor do you choose?

Answer

A vasodilator without exercise, because exercise or dobutamine can cause a false septal defect.

#59 · Selection of Nuclear Cardiology Imaging Tests

Which stressor is preferred for a patient with active wheezing who cannot take a vasodilator?

Answer

Dobutamine.

#60 · Selection of Nuclear Cardiology Imaging Tests

Which vasodilator is usually chosen for mild to moderate asthma without active wheezing?

Answer

Regadenoson.

#61 · Selection of Nuclear Cardiology Imaging Tests

How long should caffeine be held before regadenoson or adenosine?

Answer

At least 12 hours, including decaffeinated products and chocolate.

#62 · Selection of Nuclear Cardiology Imaging Tests

How long should theophylline be held before vasodilator stress?

Answer

At least 48 hours.

#63 · Selection of Nuclear Cardiology Imaging Tests

How long should oral dipyridamole be held before adenosine or regadenoson?

Answer

At least 48 hours, since it intensifies and prolongs the adenosine effect.

#64 · Selection of Nuclear Cardiology Imaging Tests

A patient drank coffee this morning before a regadenoson study. What is the risk?

Answer

A false negative study, because caffeine blocks adenosine receptors; reschedule or flag the risk.

#65 · Selection of Nuclear Cardiology Imaging Tests

List contraindications to adenosine or dipyridamole stress.

Answer

Second or third degree AV block or sinus node dysfunction without a pacemaker, active bronchospasm, systolic pressure below about 90 mmHg, and hypersensitivity.

#66 · Selection of Nuclear Cardiology Imaging Tests

What safety warning does regadenoson carry besides AV block?

Answer

A seizure risk, which matters in patients with seizure history or on drugs that lower the threshold.

#67 · Selection of Nuclear Cardiology Imaging Tests

A treadmill test ends at 70 percent of maximum predicted heart rate with no ST change or symptoms. What is the usual next step?

Answer

Convert to a vasodilator study, with the tracer given during the vasodilator.

#68 · Selection of Nuclear Cardiology Imaging Tests

How long should the patient keep exercising after tracer injection?

Answer

About 1 minute with Tc-99m agents or thallium, so uptake occurs while flow is high.

#69 · Selection of Nuclear Cardiology Imaging Tests

Why add low level exercise to a vasodilator study?

Answer

It reduces side effects, lowers liver and gut activity and improves image quality.

#70 · Selection of Nuclear Cardiology Imaging Tests

When should low level exercise NOT be combined with a vasodilator?

Answer

With LBBB or a paced rhythm.

#71 · Selection of Nuclear Cardiology Imaging Tests

What are the half-life and photon energy of Tc-99m?

Answer

6.0 hours and 140 keV.

#72 · Selection of Nuclear Cardiology Imaging Tests

What are the half-life and emissions of Tl-201?

Answer

73 hours, with mercury x-rays of 69 to 83 keV.

#73 · Selection of Nuclear Cardiology Imaging Tests

What is the first pass extraction of thallium compared with the technetium agents?

Answer

About 85 percent, versus about 60 percent for sestamibi and 50 percent for tetrofosmin.

#74 · Selection of Nuclear Cardiology Imaging Tests

What is redistribution and why is it clinically useful?

Answer

Thallium washes out of normal myocardium and refills hypoperfused but viable regions over hours, which supports viability imaging and single injection rest imaging.

#75 · Selection of Nuclear Cardiology Imaging Tests

Why do Tc-99m agents need two injections for rest and stress?

Answer

They do not redistribute appreciably, so the image shows flow at the moment of injection.

#76 · Selection of Nuclear Cardiology Imaging Tests

What is the half-life of Rb-82 and how is it supplied?

Answer

About 76 seconds, from a Sr-82 generator (half-life 25.5 days) replaced about every 4 weeks.

#77 · Selection of Nuclear Cardiology Imaging Tests

What are the Sr-82 and Sr-85 breakthrough limits for Rb-82?

Answer

0.02 microcuries of Sr-82 and 0.2 microcuries of Sr-85 per millicurie of Rb-82.

#78 · Selection of Nuclear Cardiology Imaging Tests

What is the half-life of N-13 ammonia and what does it require?

Answer

About 10 minutes, and an on-site cyclotron.

#79 · Selection of Nuclear Cardiology Imaging Tests

Why can F-18 flurpiridaz be used with exercise stress?

Answer

Its 110 minute half-life lets the tracer be injected at peak exercise, and it has the shortest positron range of the perfusion agents.

#80 · Selection of Nuclear Cardiology Imaging Tests

Is F-18 FDG a perfusion tracer?

Answer

No. It is a glucose analog used for viability and inflammation imaging.

#81 · Selection of Nuclear Cardiology Imaging Tests

What is a typical one-day Tc-99m dose pair?

Answer

About 8 to 12 mCi first and 24 to 36 mCi second, with the second at least three times the first.

#82 · Selection of Nuclear Cardiology Imaging Tests

What is the usual thallium stress dose?

Answer

2.5 to 3.5 mCi (92 to 130 MBq), with about 1 mCi at reinjection.

#83 · Selection of Nuclear Cardiology Imaging Tests

Compute the Tc-99m activity left after 6 hours from a 100 mCi start.

Answer

One half-life has passed, so 100 x 0.5 = 50 mCi.

#84 · Selection of Nuclear Cardiology Imaging Tests

How do you compute activity after a decay time t?

Answer

A = A0 x 0.5 raised to the power (t divided by half-life), using the same time units.

#85 · Selection of Nuclear Cardiology Imaging Tests

Rank the approximate effective doses of common perfusion studies.

Answer

N-13 ammonia about 2 mSv, Rb-82 about 3 to 4 mSv, Tc-99m one day about 9 to 11 mSv, and Tl-201 at 3 mCi about 14 mSv or more.

#86 · Selection of Nuclear Cardiology Imaging Tests

Which common SPECT protocol gives the highest radiation dose?

Answer

Dual isotope.

#87 · Selection of Nuclear Cardiology Imaging Tests

When is stress only imaging appropriate?

Answer

When the patient is low risk, the stress images are normal, and attenuation correction or prone imaging makes a normal result reliable.

#88 · Selection of Nuclear Cardiology Imaging Tests

When is stress only a poor choice?

Answer

With known prior infarction, a likely fixed defect, or a viability question.

#89 · Selection of Nuclear Cardiology Imaging Tests

What is the usual sequence in thallium stress protocols?

Answer

Stress first, redistribution imaging at 3 to 4 hours, then optional reinjection and a 24 hour image.

#90 · Selection of Nuclear Cardiology Imaging Tests

What are the drawbacks of dual isotope imaging?

Answer

Higher dose, technetium down scatter into the thallium window and poorer thallium image quality.

#91 · Selection of Nuclear Cardiology Imaging Tests

What patient preparation does FDG viability imaging need?

Answer

A glucose load or hyperinsulinemic euglycemic clamp so the heart uses glucose, with careful insulin management in diabetics.

#92 · Selection of Nuclear Cardiology Imaging Tests

What does a perfusion and FDG mismatch mean?

Answer

Reduced perfusion with preserved FDG uptake, indicating hibernating viable myocardium.

#93 · Selection of Nuclear Cardiology Imaging Tests

What is myocardial flow reserve and what value is commonly abnormal?

Answer

Stress flow divided by rest flow; below about 2.0 is commonly abnormal.

#94 · Selection of Nuclear Cardiology Imaging Tests

What are the main PET advantages over SPECT?

Answer

Higher resolution, reliable attenuation correction, fewer equivocal studies, lower dose, speed and absolute flow quantitation.

#95 · Selection of Nuclear Cardiology Imaging Tests

What is the best test for serial, reproducible LVEF during cardiotoxic chemotherapy?

Answer

Equilibrium radionuclide ventriculography (MUGA) with Tc-99m labeled red cells.

#96 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How long before a vasodilator stress study should caffeine be held?

Answer

At least 12 hours, including coffee, tea, cola and energy drinks. Decaffeinated coffee still contains some caffeine.

#97 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How long is theophylline held before a vasodilator study when the clinical situation allows?

Answer

About 48 hours, because it is a methylxanthine that blocks adenosine receptors.

#98 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why must dipyridamole be held for at least 24 hours before an adenosine study?

Answer

Dipyridamole blocks adenosine reuptake and potentiates adenosine, which can cause profound bradycardia or heart block at standard adenosine doses.

#99 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the target heart rate for exercise or dobutamine stress?

Answer

At least 85 percent of the age-predicted maximum heart rate, where the maximum is estimated as 220 minus age.

#100 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Calculate the 85 percent target heart rate for a 58-year-old.

Answer

Maximum is 220 - 58 = 162; 0.85 x 162 = 137.7, so about 138 beats per minute.

#101 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A patient on a beta blocker reaches only 77 percent of predicted maximum heart rate. What do you conclude?

Answer

The test is submaximal and a negative result has reduced sensitivity. Add or convert to a vasodilator stress rather than calling it negative.

#102 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why are beta blockers usually held 24 to 48 hours before a diagnostic exercise or dobutamine study?

Answer

They blunt the heart rate and rate pressure product, so the patient may not reach target and ischemia may not be provoked.

#103 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Which resting-ECG drug effect makes exercise ST changes uninterpretable?

Answer

Digoxin, which can cause ST depression that mimics ischemia.

#104 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What are the first four stages of the Bruce protocol?

Answer

Each lasts 3 minutes: 1.7 mph at 10 percent, 2.5 mph at 12 percent, 3.4 mph at 14 percent, 4.2 mph at 16 percent.

#105 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

When is the radiotracer injected during exercise stress, and how long does exercise continue?

Answer

At or near peak exercise, with exercise continuing about 1 minute after injection.

#106 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Write the Duke treadmill score formula.

Answer

Exercise time in minutes - (5 x maximum ST deviation in mm) - (4 x angina index), with angina index 0 none, 1 non-limiting, 2 exercise-limiting.

#107 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Compute the Duke score for 9 minutes, 2 mm ST depression and limiting angina.

Answer

9 - 10 - 8 = -9, which is intermediate risk.

#108 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What are the Duke score risk categories?

Answer

Low risk +5 or higher, intermediate -10 to +4, high risk -11 or lower.

#109 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Name five absolute contraindications to exercise stress.

Answer

Acute MI within 2 days, high-risk unstable angina, uncontrolled symptomatic arrhythmia, symptomatic severe aortic stenosis, decompensated heart failure; also acute PE, acute myocarditis or pericarditis, acute aortic dissection.

#110 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Give examples of relative contraindications to exercise testing.

Answer

Left main stenosis, moderate valve stenosis, electrolyte abnormality, severe hypertension (about 200/110 mm Hg or higher), high-degree AV block, obstructive hypertrophic cardiomyopathy.

#111 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Name absolute indications to terminate exercise.

Answer

More than 10 mm Hg systolic drop with ischemia signs, moderate to severe angina, neurologic symptoms, poor perfusion signs, sustained VT, ST elevation above 1 mm in leads without Q waves, equipment failure, or patient request.

#112 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Is more than 2 mm of horizontal ST depression an absolute or relative reason to stop exercise?

Answer

Relative. It is a relative indication to stop, along with fatigue, wheezing, claudication, and a nonischemic systolic drop.

#113 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A patient with left bundle branch block needs stress perfusion imaging. Which stress is preferred?

Answer

A vasodilator, because exercise or fast heart rates can cause a septal defect from abnormal contraction timing, not coronary disease.

#114 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why is low-level exercise added to vasodilator stress, and when is it avoided?

Answer

It reduces side effects and subdiaphragmatic tracer activity. It is avoided in left bundle branch block.

#115 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What follows the regadenoson bolus, and when is the radiotracer injected?

Answer

A 5 mL saline flush follows the bolus immediately, and the radiotracer is injected about 10 to 20 seconds after the flush.

#116 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

When is the tracer injected after regadenoson?

Answer

10 to 20 seconds after the saline flush.

#117 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Which adenosine receptor does regadenoson target selectively, and what is the benefit?

Answer

The A2A receptor, so it causes less atrioventricular block and bronchospasm than adenosine.

#118 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Which adenosine receptor subtypes cause bronchoconstriction and AV node slowing?

Answer

A2B and A3 cause bronchoconstriction; A1 slows the sinoatrial and atrioventricular nodes.

#119 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the standard adenosine dose and duration?

Answer

140 mcg/kg/min intravenous infusion over 6 minutes, or over 4 minutes in the abbreviated protocol.

#120 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

When is the tracer injected during a 6-minute adenosine infusion?

Answer

At 3 minutes (the midpoint); in the 4-minute protocol, at 2 minutes. Infusion continues 2 to 3 minutes after injection.

#121 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Calculate the adenosine infusion rate and 6-minute total for an 80 kg patient.

Answer

80 x 140 = 11,200 mcg/min = 11.2 mg/min; 11.2 x 6 = 67.2 mg total.

#122 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the half-life of adenosine and what does it mean clinically?

Answer

Under 10 seconds, so side effects usually end within a minute or two of stopping the infusion.

#123 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the dipyridamole dose and when is the tracer injected?

Answer

0.142 mg/kg/min over 4 minutes (0.568 mg/kg total), with tracer 3 to 5 minutes after the infusion ends.

#124 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Calculate the dipyridamole total dose for a 75 kg patient.

Answer

0.568 x 75 = 42.6 mg (10.65 mg/min for 4 minutes).

#125 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why may dipyridamole side effects return after improving?

Answer

Its effect is long because it raises endogenous adenosine, so aminophylline must be available and may be needed again.

#126 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

List the vasodilator contraindications.

Answer

Second or third degree AV block or sinus node dysfunction without a pacemaker, active bronchospasm for adenosine and dipyridamole, systolic pressure under about 90 mm Hg, and recent caffeine or theophylline.

#127 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Describe the dobutamine stress protocol.

Answer

Start 5 mcg/kg/min, increase every 3 minutes to 10, 20, 30 and 40 mcg/kg/min; add atropine 0.25 mg increments up to about 2 mg if target heart rate is not reached.

#128 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Calculate the dobutamine rate for a 70 kg patient at 20 mcg/kg/min.

Answer

70 x 20 = 1,400 mcg/min = 1.4 mg/min.

#129 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

When is dobutamine chosen over a vasodilator?

Answer

When vasodilators are contraindicated, such as active bronchospasm or recent methylxanthine use.

#130 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Name contraindications to dobutamine stress.

Answer

Significant ventricular arrhythmia, severe uncontrolled hypertension, aortic dissection, severe aortic stenosis, obstructive hypertrophic cardiomyopathy, and LV thrombus.

#131 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the typical aminophylline reversal dose?

Answer

50 to 250 mg by slow intravenous injection over 30 to 60 seconds, given at least 1 to 2 minutes after the tracer.

#132 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A patient has a seizure after regadenoson. What do you give?

Answer

Supportive care, airway protection and a benzodiazepine. Avoid aminophylline because it lowers the seizure threshold.

#133 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A patient develops wheezing during adenosine. What do you do first?

Answer

Stop the infusion and give inhaled albuterol; add aminophylline if wheezing persists.

#134 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What do you do for hypotension during vasodilator stress?

Answer

Stop the agent, lay the patient supine with legs raised, give fluids, and add aminophylline if it does not recover.

#135 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What reverses dobutamine-induced ischemia or tachyarrhythmia?

Answer

Intravenous esmolol or metoprolol.

#136 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the first step for almost any stress complication?

Answer

Stop the stress agent, then treat the specific problem.

#137 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What are the signs and the management of tracer extravasation?

Answer

Swelling, pain, resistance to the flush, or a hot spot at the site on the image. Stop, elevate the arm, apply a warm compress, document it, notify the physician, and reinject if needed.

#138 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

When does imaging start after Tl-201 stress injection, and when are redistribution images taken?

Answer

Within about 10 to 15 minutes of injection, with redistribution images 3 to 4 hours later.

#139 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why is Tc-99m sestamibi imaging delayed longer after vasodilator injection than after exercise?

Answer

Liver and gut activity is higher after vasodilator, so about 45 to 60 minutes is needed versus about 15 to 30 minutes after exercise.

#140 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

In a one-day Tc-99m rest-stress protocol, how do the doses compare?

Answer

The stress dose is about 3 times the rest dose to override residual rest activity, for example rest 8 to 12 mCi and stress 24 to 36 mCi.

#141 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What material forms the scintillation crystal of a standard gamma camera?

Answer

Thallium-doped sodium iodide, NaI(Tl). It converts the photon's energy into visible light. A cardiac crystal is typically 3/8 inch (about 9.5 mm) thick.

#142 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does the total pulse height from the photomultiplier tubes represent?

Answer

The energy deposited by the photon. This is how the pulse height analyzer separates primary photons from scattered ones.

#143 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How does the camera determine where a photon struck the crystal?

Answer

From the relative signal strength in neighboring photomultiplier tubes (Anger positioning logic).

#144 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Compute the 20 percent energy window for Tc-99m.

Answer

10 percent of 140 keV is 14 keV, so the window runs from 126 to 154 keV, centered on the 140 keV photopeak.

#145 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Which Tl-201 emissions are used for imaging and what windows are typical?

Answer

Mercury x-rays at about 70 to 80 keV (a 30 percent window on 70 keV is common) plus the 167 keV gamma ray (a 20 percent window).

#146 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why must Tl-201 rest images be acquired before Tc-99m stress injection in a dual isotope study?

Answer

Higher-energy Tc-99m photons scatter down into the Tl-201 window and contaminate it, so the Tl-201 study is done first.

#147 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the trade-off between LEHR and LEGP collimators?

Answer

LEHR has longer, narrower holes: better resolution, fewer counts. LEGP has shorter, wider holes: more counts, worse resolution.

#148 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What happens to spatial resolution as the heart moves farther from the collimator face?

Answer

It gets worse. Keep the detector as close to the chest as possible, using an elliptical or body-contour orbit when available.

#149 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does a photon that scatters in the patient do to the image if it is accepted?

Answer

It carries wrong position information because it changed direction and lost energy, which blurs the image and lowers contrast.

#150 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What are the effects of an energy window set too wide?

Answer

It accepts more scattered photons, lowering image contrast. A window shifted off the photopeak lowers counts and degrades uniformity.

#151 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How does a CZT detector differ from a NaI(Tl) camera at the physics level?

Answer

CZT converts photons directly to an electrical signal, with no scintillation crystal and no photomultiplier tubes.

#152 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Compare energy resolution of CZT and NaI(Tl) at 140 keV.

Answer

CZT is about 5 to 6 percent. NaI(Tl) is roughly 9 to 10 percent. Better energy resolution means better scatter rejection.

#153 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What practical advantages does the high sensitivity of a dedicated cardiac CZT camera give?

Answer

Acquisition in a few minutes, a reduced radiotracer dose at normal time, or a mix of the two.

#154 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the standard orbit for conventional Tc-99m cardiac SPECT?

Answer

180 degrees, from 45 degrees right anterior oblique to 45 degrees left posterior oblique, because the heart lies in the left anterior chest.

#155 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

In a dual-head camera in L mode, how far does each head rotate to cover a 180 degree orbit?

Answer

90 degrees each, because the heads sit 90 degrees apart, so the arc is covered in half the time.

#156 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What are typical matrix and pixel size for cardiac SPECT?

Answer

A 64 x 64 matrix with a pixel size of about 6 to 7 mm.

#157 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why does moving from a 64 x 64 to a 128 x 128 matrix increase noise at the same counts?

Answer

The same counts are divided among four times as many pixels, so each pixel has fewer counts and more statistical noise.

#158 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A lab uses 25 seconds per projection at 8 mCi. What time keeps the same counts at 4 mCi?

Answer

50 seconds. The dose is halved so time doubles: 8 x 25 equals 4 x 50, both 200.

#159 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the downside of lowering dose and lengthening acquisition?

Answer

The longer study raises the chance of patient motion artifact.

#160 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What are typical ASNC one-day Tc-99m doses for the first and second injections?

Answer

About 8 to 10 mCi (296 to 370 MBq) first, then about 25 to 30 mCi (925 to 1,110 MBq), so the second dose overwhelms residual activity from the first.

#161 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Typical delay between Tc-99m sestamibi or tetrofosmin injection and imaging?

Answer

About 15 to 30 minutes after exercise, and about 45 to 60 minutes at rest or after vasodilator stress without exercise.

#162 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Which defect pattern does breast attenuation create and how do you reduce it?

Answer

Lowered anterior and anterolateral counts. Position the breast consistently between rest and stress, and consider attenuation correction.

#163 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the benefit and a risk of prone imaging?

Answer

It moves the diaphragm and abdominal organs away from the inferior wall and often removes inferior attenuation. It can create an apparent anterior defect, so it is usually a second set rather than a replacement.

#164 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A SPECT/CT study shows a new anterior wall defect on corrected images only. What is the first thing to check?

Answer

CT to emission registration. If the heart on CT overlies lung, the correction under-compensates and creates a false defect. Realign and re-review.

#165 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why should attenuation-corrected and uncorrected images both be reviewed?

Answer

Correction can create its own artifacts, and agreement between the two strengthens confidence in a finding.

#166 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How do dual or triple energy window methods correct scatter?

Answer

They sample counts in adjacent windows to estimate the scatter distribution under the photopeak and subtract it.

#167 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does resolution recovery correct?

Answer

The distance-dependent blur of the collimator, modeled inside iterative reconstruction. It can allow lower dose or shorter time.

#168 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What trigger point does ECG gating use to bin counts?

Answer

The R wave. Counts are assigned to frames starting at each R wave.

#169 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How many frames per cardiac cycle are standard, and what does 16 frames change?

Answer

Eight is standard. Sixteen gives a more accurate ejection fraction and diastolic assessment but fewer counts per frame.

#170 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is an R-R acceptance window?

Answer

A range around the average R-R interval, commonly plus or minus 20 to 30 percent. A beat outside it is rejected as a bad beat.

#171 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What gating problem do frequent PVCs and atrial fibrillation cause?

Answer

Variable R-R intervals starve the last frames of counts (end-diastolic dropout) and many rejected beats lengthen the study and lower counts.

#172 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What do you do when gating is poor because of arrhythmia?

Answer

Review the rejection rate, widen the window moderately, or use list mode if available. If still poor, report perfusion and state that the ejection fraction is unreliable.

#173 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Calculate EF with EDV 120 mL and ESV 50 mL.

Answer

Stroke volume is 70 mL, and 70 divided by 120 equals 0.583, so EF is about 58 percent.

#174 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why can gated SPECT overestimate EF in a small heart?

Answer

The coarse matrix underestimates small cavity volumes, which inflates the apparent EF.

#175 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does the ramp filter do in filtered back projection?

Answer

It removes the blur of simple back projection but amplifies high-frequency noise, so a low pass filter is added.

#176 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the Nyquist frequency in cycles per pixel?

Answer

0.5 cycles per pixel, the highest spatial frequency the pixel grid can represent.

#177 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What happens when a Butterworth cutoff is set too low?

Answer

Images are very smooth but resolution and defect contrast drop, so small or subtle defects may be blurred away.

#178 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What happens when a Butterworth cutoff is set too high?

Answer

More detail is kept but more noise passes, so image noise rises, especially in low count studies.

#179 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why can intense liver or bowel activity create an inferior wall defect with filtered back projection?

Answer

The ramp filter produces negative lobes next to hot activity, lowering counts in the adjacent inferior wall. Iterative reconstruction handles this less severely.

#180 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does OSEM stand for and why use subsets?

Answer

Ordered subsets expectation maximization. Grouping projections into subsets speeds convergence compared with MLEM.

#181 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is the problem with too many iterations in iterative reconstruction?

Answer

Noise is amplified, which is why a post reconstruction smoothing filter is used.

#182 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why can quantitative results from two different reconstruction settings not be compared?

Answer

Noise texture and count behavior differ by method, so normal databases and thresholds must match the reconstruction used.

#183 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How is the left ventricular long axis defined for reorientation?

Answer

A line from the apex through the center of the mitral valve plane.

#184 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Which walls does the vertical long axis show, and which does the horizontal long axis show?

Answer

Vertical long axis: anterior, apical and inferior walls. Horizontal long axis: septal, apical and lateral walls.

#185 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A polar map shows an odd defect but the short axis stack looks normal. What should you check first?

Answer

Reorientation and long axis placement. A wrong axis can create a false apical or opposite wall defect, so also check the mask and valve plane.

#186 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What is upward creep and when is it likely?

Answer

Gradual rise of the heart in the chest as breathing settles, causing a vertical shift across the orbit. It is more likely when imaging starts too soon after exercise.

#187 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How do you detect motion before reading a SPECT study?

Answer

Review the raw projection cine and the sinogram or linogram. A step or discontinuity indicates motion.

#188 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What do you do about large patient motion?

Answer

Repeat the acquisition. Software realignment fixes a small shift but cannot rescue large or complex movement.

#189 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What artifact does a nonuniform detector produce in SPECT, and what QC test finds it?

Answer

A ring artifact, because the same defect appears in every projection. The daily flood field uniformity check finds it.

#190 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

In what order should you review a SPECT study to avoid missing an artifact source?

Answer

Raw projections first (motion, extracardiac uptake, counts, gating), then slices, polar maps and quantitation, gated function, and CT registration last.

#191 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How does an intrinsic flood differ from an extrinsic flood?

Answer

Intrinsic: collimator removed, point source far from the crystal. Extrinsic: collimator on, uniform sheet or filled phantom on the collimator.

#192 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does an extrinsic flood detect that an intrinsic flood cannot?

Answer

Collimator damage, because only the extrinsic flood is acquired through the collimator.

#193 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A flood shows a round cold spot about the size of one tube. What is the likely cause?

Answer

A photomultiplier tube that has drifted low in gain or failed.

#194 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A flood shows a thin linear cold line that cannot be retuned away. What is the likely cause?

Answer

A cracked crystal, often from thermal shock; it cannot be repaired.

#195 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A cold line appears on the extrinsic flood but not on the intrinsic flood. Where is the fault?

Answer

In the collimator, not the crystal or electronics.

#196 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What energy window setup is used for Tc-99m peaking?

Answer

Center on 140 keV with a window of about 15 to 20 percent.

#197 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How far should a point source be from the crystal for an intrinsic flood?

Answer

At least five times the largest dimension of the detector, so the photons arrive nearly uniformly.

#198 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Compute integral uniformity when the hottest pixel has 2,600 counts and the coldest 2,300.

Answer

100 times (2,600 minus 2,300) divided by (2,600 plus 2,300) = 100 times 300 divided by 4,900 = 6.1 percent.

#199 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does a center of rotation error look like in SPECT slices?

Answer

A ring or tuning fork shape around point sources, with blurring of the myocardium.

#200 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does the COR test acquire and fit?

Answer

An off-center point source over a full orbit; its position in each projection is fit to a sine wave and the deviation is the error.

#201 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Which solid-state camera flood pattern replaces the PMT-out pattern?

Answer

Bad pixels or detector modules, since CZT cameras have no photomultiplier tubes.

#202 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Which agency regulation covers dose calibrator and survey meter QC?

Answer

The NRC, in 10 CFR 35.60 for dose calibrators and 35.61 for survey meters.

#203 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Does the NRC prescribe a specific daily camera flood test?

Answer

No. Camera QC follows the manufacturer, society guidance, the accrediting body and facility procedures.

#204 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How often is dose calibrator constancy tested and what is the tolerance?

Answer

Every day of use, within 10 percent of the decay-corrected expected reading of a long-lived source such as Cs-137.

#205 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How often is dose calibrator accuracy tested?

Answer

At installation, annually and after repair, using at least two traceable sources of different energies, within 10 percent.

#206 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How often is dose calibrator linearity tested and over what range?

Answer

At installation, quarterly and after repair, from the highest activity used down to about 10 microcuries, within 10 percent.

#207 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

When is dose calibrator geometry tested?

Answer

At installation and after relocation or repair, showing the reading does not depend on sample volume or position.

#208 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A dose calibrator is moved to a new room. Which tests are repeated?

Answer

Geometry and accuracy, in addition to the usual schedule.

#209 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Name two ways to test dose calibrator linearity.

Answer

The decay method using Tc-99m measured over several days, or calibrated lead sleeves (attenuators).

#210 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

The expected Cs-137 constancy reading is 198 microcuries and the measured value is 215. Does it pass?

Answer

Difference 17 divided by 198 is 8.6 percent, within the 10 percent tolerance; it passes but warrants a repeat and a trend check.

#211 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How often is a survey meter calibrated and what is the tolerance?

Answer

At least annually and after repair, at two points on each scale, within 20 percent of the true value.

#212 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What daily test checks the Rb-82 generator?

Answer

Elution testing for breakthrough of Sr-82 and Sr-85, because Sr-82 is bone seeking and adds dose without image value.

#213 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What are the label limits for Rb-82 generator breakthrough?

Answer

0.02 microcurie Sr-82 and 0.2 microcurie Sr-85 per millicurie of Rb-82; reaching them retires the generator.

#214 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why is PET scanner cross-calibration with the dose calibrator important?

Answer

A calibration error carries directly into absolute flow values in mL per minute per g.

#215 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does a daily PET blank scan reveal?

Answer

A failed detector block, seen as a streak in the sinogram compared with baseline.

#216 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Compare Rb-82 and N-13 ammonia for spatial resolution and extraction.

Answer

N-13 ammonia has a shorter positron range and better extraction at high flow, so better resolution; Rb-82 has a longer range and flow roll-off.

#217 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What are the half-lives of Rb-82, N-13 and F-18 flurpiridaz?

Answer

Rb-82 about 76 seconds, N-13 about 10 minutes, flurpiridaz about 110 minutes.

#218 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why can F-18 flurpiridaz be used without an on-site cyclotron and with exercise stress?

Answer

Its half-life of about 110 minutes allows delivery and imaging after exercise.

#219 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

When does static Rb-82 imaging begin?

Answer

After blood pool clearance, about 90 seconds after infusion start with a normal LVEF and later when LVEF is low.

#220 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

When should list mode acquisition start for PET flow quantification?

Answer

At the time of tracer injection.

#221 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Define myocardial flow reserve and compute it for rest 0.9 and stress 2.1.

Answer

Stress MBF divided by rest MBF; 2.1 divided by 0.9 is 2.33.

#222 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Rest MBF is 1.2 at RPP 12,000 and stress MBF is 1.8. What is the corrected MFR for RPP 10,000?

Answer

Corrected rest is 1.2 times 10,000 divided by 12,000 = 1.0, so MFR is 1.8 divided by 1.0 = 1.8 (1.5 uncorrected).

#223 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why can a low MFR occur with a normal relative perfusion image?

Answer

Relative images only compare regions with the best region, so balanced multivessel or microvascular disease can look normal while absolute flow reserve is low.

#224 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Name two nondisease reasons for a falsely low MFR.

Answer

A high resting flow from high cardiac work, and a blunted stress response from inadequate vasodilation or caffeine.

#225 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How are diabetic patients prepared for FDG viability imaging?

Answer

Hyperinsulinemic euglycemic clamp, or titrated IV insulin with glucose, checking blood glucose before FDG.

#226 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What does FDG uptake with reduced perfusion mean?

Answer

A mismatch, indicating hibernating viable myocardium; a match (both reduced) indicates scar.

#227 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

A defect appears only on CT attenuation-corrected images and the heart overlaps lung on the CT. What do you do?

Answer

Suspect misregistration, realign the CT to the emission data and reprocess, comparing with uncorrected images.

#228 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How does a heart projecting over lung on the CT change the corrected image?

Answer

It is under-corrected, producing an artifactual anterior or lateral defect.

#229 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How are Agatston density factors assigned?

Answer

130 to 199 HU factor 1, 200 to 299 factor 2, 300 to 399 factor 3, 400 or more factor 4, multiplied by lesion area.

#230 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Compute an Agatston score: 6 mm squared at 180 HU, 10 mm squared at 320 HU, 4 mm squared at 450 HU.

Answer

6 times 1 plus 10 times 3 plus 4 times 4 = 6 plus 30 plus 16 = 52.

#231 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

Why is the attenuation correction CT not used for an Agatston score?

Answer

It is ungated, so motion makes calcium quantification unreliable; calcium needs a gated non-contrast CT.

#232 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How is a patient prepared for cardiac sarcoid FDG imaging?

Answer

High fat, low carbohydrate diet for at least 24 hours, then a fast of at least 4 to 12 hours, with optional IV heparin about 50 units per kg.

#233 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

How is cardiac Tc-99m PYP graded and which grades are positive?

Answer

Heart versus rib: 0 none, 1 less than rib, 2 equal, 3 greater; grades 2 and 3 are positive, and SPECT is required.

#234 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What must be excluded after a positive PYP scan, and how?

Answer

AL amyloid, using serum free light chains and serum and urine immunofixation; a monoclonal protein leads to biopsy.

#235 · Performance of Imaging Tests Including Instrumentation Protocols and Processing

What are the I-123 MIBG preparation, imaging times and key ratio threshold?

Answer

Thyroid blockade first; planar images at 15 minutes and 4 hours; a late heart to mediastinum ratio below 1.6 is abnormal.

#236 · Interpretation of Nuclear Cardiology Imaging Tests

What is the first step in reading any SPECT MPI study?

Answer

Review the raw projection data (cine, sinogram) before the reconstructed slices. Raw review shows motion, attenuation, extracardiac activity and comparability.

#237 · Interpretation of Nuclear Cardiology Imaging Tests

How does a sinogram display one slice of raw data?

Answer

Projection angle runs on the vertical axis and detector position on the horizontal axis. A stationary point of activity traces a smooth sine wave.

#238 · Interpretation of Nuclear Cardiology Imaging Tests

How does patient motion appear on a sinogram?

Answer

As a break, kink or step in the otherwise smooth sine wave at the moment of motion. A vertical shift moves the whole band up or down.

#239 · Interpretation of Nuclear Cardiology Imaging Tests

What does a linogram show?

Answer

A redisplay in which stationary activity falls on a straight line. Motion shows as a break in the line.

#240 · Interpretation of Nuclear Cardiology Imaging Tests

What is upward creep?

Answer

A progressive rise of the heart in the chest during acquisition as ventilation and the diaphragm settle after exercise. It creates artifactual defects.

#241 · Interpretation of Nuclear Cardiology Imaging Tests

How do you prevent upward creep?

Answer

Delay the start of imaging a few more minutes after exercise, or repeat the acquisition once the patient has settled. Cine shows the heart rising against a fixed background.

#242 · Interpretation of Nuclear Cardiology Imaging Tests

What is the typical pattern of breast attenuation?

Answer

A fixed defect in the anterior wall and anterior septum, sometimes the apex, with normal thickening on gated images.

#243 · Interpretation of Nuclear Cardiology Imaging Tests

A woman has a fixed anterior defect with normal wall thickening and a shadow on cine. What is the likely cause?

Answer

Breast attenuation artifact. Normal gated thickening argues against infarct.

#244 · Interpretation of Nuclear Cardiology Imaging Tests

Why must breast position be the same on stress and rest?

Answer

A shifted breast changes the attenuation shadow and can create an artifactual reversible defect that mimics ischemia.

#245 · Interpretation of Nuclear Cardiology Imaging Tests

How can breast attenuation be reduced?

Answer

Position or strap the breast away from the heart in the same way for both sets, and consider attenuation correction.

#246 · Interpretation of Nuclear Cardiology Imaging Tests

What is the typical pattern of diaphragm attenuation?

Answer

Reduced counts in the inferior wall, more common in men and large patients, with normal inferior thickening on gating.

#247 · Interpretation of Nuclear Cardiology Imaging Tests

What is the best fix for a suspected diaphragm attenuation defect?

Answer

Prone imaging, which moves the diaphragm and abdominal contents away from the inferior wall.

#248 · Interpretation of Nuclear Cardiology Imaging Tests

What artifact can prone imaging introduce?

Answer

An artifactual anterior or anteroseptal defect from chest compression against the table. Interpret supine and prone together.

#249 · Interpretation of Nuclear Cardiology Imaging Tests

What does upright imaging do compared with supine imaging?

Answer

It lowers the diaphragm and can reduce inferior attenuation. Its role is similar to prone imaging.

#250 · Interpretation of Nuclear Cardiology Imaging Tests

Why are arms raised overhead during acquisition?

Answer

It removes the arms from the lateral photon path and reduces attenuation. Arm position should match on stress and rest.

#251 · Interpretation of Nuclear Cardiology Imaging Tests

What causes the ramp filter artifact?

Answer

Filtered back projection uses a ramp filter that makes negative lobes around intense activity. The adjacent wall is assigned too few counts.

#252 · Interpretation of Nuclear Cardiology Imaging Tests

A bright bowel loop sits beside a dim inferior wall on a sestamibi study. What is the best next step?

Answer

Have the patient drink water or wait for gut clearance, then reacquire. Do not report an inferior perfusion defect yet.

#253 · Interpretation of Nuclear Cardiology Imaging Tests

How much water is given to help clear subdiaphragmatic activity?

Answer

About one to two glasses (8 to 16 ounces), then wait and reimage. This helps empty the stomach and duodenum.

#254 · Interpretation of Nuclear Cardiology Imaging Tests

Roughly how long after injection is rest sestamibi imaging usually started?

Answer

About 45 to 60 minutes. After exercise stress, about 15 to 30 minutes, with a longer wait after vasodilator stress.

#255 · Interpretation of Nuclear Cardiology Imaging Tests

Does low level exercise help during vasodilator stress?

Answer

Yes. It reduces gut and liver activity compared with vasodilator stress alone.

#256 · Interpretation of Nuclear Cardiology Imaging Tests

Why does intense extracardiac activity cause problems besides the ramp filter?

Answer

Scatter and overlap add apparent counts to the adjacent wall and can hide a true inferior defect.

#257 · Interpretation of Nuclear Cardiology Imaging Tests

How does a large pericardial effusion look on raw cine?

Answer

As a photopenic halo around the heart, and the heart may seem to swing. Correlate with echocardiography.

#258 · Interpretation of Nuclear Cardiology Imaging Tests

Which tracers can accumulate in soft tissue tumors?

Answer

Thallium-201 and Tc-99m sestamibi can accumulate in some lung, breast, thyroid and lymphoma lesions. Report focal uptake and correlate with CT.

#259 · Interpretation of Nuclear Cardiology Imaging Tests

How can a hiatal hernia mimic a cardiac abnormality?

Answer

It lies behind the heart and can take up tracer, mimicking an inferior defect or hot area. Water and reimaging help separate stomach from myocardium.

#260 · Interpretation of Nuclear Cardiology Imaging Tests

What does increased lung uptake of thallium after stress indicate?

Answer

Raised left atrial and pulmonary capillary pressure, marking LV dysfunction, extensive coronary disease and worse prognosis.

#261 · Interpretation of Nuclear Cardiology Imaging Tests

What lung-to-heart ratio is abnormal on planar thallium imaging?

Answer

A ratio above about 0.5 is considered abnormal.

#262 · Interpretation of Nuclear Cardiology Imaging Tests

What is the normal RV uptake relative to the LV?

Answer

The RV is faint and clearly less intense than the LV.

#263 · Interpretation of Nuclear Cardiology Imaging Tests

What does RV uptake equal to LV uptake suggest?

Answer

Right ventricular hypertrophy from pulmonary hypertension or lung disease. A dilated RV with a flattened septum suggests overload.

#264 · Interpretation of Nuclear Cardiology Imaging Tests

A small bright spot inside the LV cavity on slices is most likely what?

Answer

Papillary muscle uptake, a normal finding that can mimic a defect or hot spot.

#265 · Interpretation of Nuclear Cardiology Imaging Tests

List what must match between stress and rest sets.

Answer

Patient and arm position, breast position, table height and orbit, and processing (orientation, filter, slices).

#266 · Interpretation of Nuclear Cardiology Imaging Tests

What happens if stress and rest slices are reoriented at different angles?

Answer

The slices do not match and a purely geometric stress to rest difference appears, especially at the apex, lateral and inferior walls. Reprocess both sets the same way.

#267 · Interpretation of Nuclear Cardiology Imaging Tests

How does center of rotation error appear?

Answer

A point source reconstructs as a ring or arc and the myocardium looks blurred or tuning-fork shaped. Suspect a quality control failure if many studies show it.

#268 · Interpretation of Nuclear Cardiology Imaging Tests

What is a limitation of filtered back projection?

Answer

Streaks and ramp filter artifact near hot structures, with noise or blur depending on the filter cutoff.

#269 · Interpretation of Nuclear Cardiology Imaging Tests

What does OSEM iterative reconstruction improve compared with filtered back projection?

Answer

It reduces streaks and negative lobes, and with resolution recovery and scatter correction it improves contrast.

#270 · Interpretation of Nuclear Cardiology Imaging Tests

What artifact can attenuation correction create?

Answer

Misregistration of the emission data and CT map causes artifactual anterior or lateral defects where the heart overlaps lung on the map. Check alignment.

#271 · Interpretation of Nuclear Cardiology Imaging Tests

Name signs of poor ECG gating.

Answer

Irregular rhythm such as atrial fibrillation or frequent premature beats, variable heart rate, poor R wave triggering and low counts.

#272 · Interpretation of Nuclear Cardiology Imaging Tests

What does the gating acceptance window do?

Answer

It keeps only beats near the average R to R interval and rejects the rest. Too narrow loses counts, too wide blurs and tends to underestimate EF.

#273 · Interpretation of Nuclear Cardiology Imaging Tests

How many frames per cycle are used in gated SPECT, and what is the tradeoff?

Answer

Eight or 16 frames. Sixteen give a more accurate EF but need more counts per frame.

#274 · Interpretation of Nuclear Cardiology Imaging Tests

What should you do with EF from a poorly gated study?

Answer

Do not rely on it. State the gating limitation and report perfusion.

#275 · Interpretation of Nuclear Cardiology Imaging Tests

Order the steps when deciding to repeat an image.

Answer

Fix gut activity (water, wait), then position (prone, upright, strap), then motion (correct or reacquire), then processing and gating, and stop for camera faults.

#276 · Interpretation of Nuclear Cardiology Imaging Tests

A study shows motion after correction and you used Tc-99m. What is the best action?

Answer

Reacquire if the motion remains significant, since Tc-99m allows a repeat the same day. Do not report a doubtful defect.

#277 · Interpretation of Nuclear Cardiology Imaging Tests

Why is repeat imaging more limited with thallium?

Answer

Thallium redistributes over time, so a later image reflects a different distribution than the initial one.

#278 · Interpretation of Nuclear Cardiology Imaging Tests

Calculate the fraction of Tc-99m activity remaining at 90 minutes (half life 360 minutes).

Answer

0.5 to the power of 90 divided by 360, which is 0.5 to the power of 0.25, equal to 0.841 or about 84 percent.

#279 · Interpretation of Nuclear Cardiology Imaging Tests

A patient had 8 mCi injected. How much remains at 90 minutes?

Answer

8 times 0.841 is about 6.7 mCi (about 249 MBq). Waiting for clearance costs little count loss.

#280 · Interpretation of Nuclear Cardiology Imaging Tests

Which defect location is helped by prone imaging, and which can it create?

Answer

It helps inferior attenuation defects and can create anterior defects from chest compression.

#281 · Interpretation of Nuclear Cardiology Imaging Tests

How many segments are in the ASNC left ventricular model, and how are they distributed?

Answer

17 segments: 6 basal, 6 mid, 4 distal, plus the apical cap as segment 17.

#282 · Interpretation of Nuclear Cardiology Imaging Tests

What is the 0 to 4 segment scoring scale?

Answer

0 normal, 1 mildly reduced, 2 moderately reduced, 3 severely reduced, 4 absent uptake.

#283 · Interpretation of Nuclear Cardiology Imaging Tests

What is the maximum possible summed stress score?

Answer

68, which is 17 segments multiplied by a top score of 4.

#284 · Interpretation of Nuclear Cardiology Imaging Tests

How is the summed difference score calculated and what does it represent?

Answer

SDS = SSS minus SRS. It measures the amount of reversible ischemia.

#285 · Interpretation of Nuclear Cardiology Imaging Tests

Stress scores sum to 11 and rest scores sum to 4. What are SDS and the percent ischemic myocardium?

Answer

SDS = 11 - 4 = 7. Percent ischemic = 7 / 68 = about 10 percent.

#286 · Interpretation of Nuclear Cardiology Imaging Tests

What SSS ranges are commonly used for normal, mild, moderate and severe abnormality?

Answer

0 to 3 normal, 4 to 8 mild, 9 to 13 moderate, above 13 severe.

#287 · Interpretation of Nuclear Cardiology Imaging Tests

Which wall and segments does the LAD usually supply?

Answer

The anterior wall, anterior septum, apex and apical cap (segments 1, 2, 7, 8, 13, 14, 17).

#288 · Interpretation of Nuclear Cardiology Imaging Tests

Which walls does the left circumflex usually supply in a right-dominant system?

Answer

The anterolateral and inferolateral walls (segments 5, 6, 11, 12, 16).

#289 · Interpretation of Nuclear Cardiology Imaging Tests

Which walls does the RCA usually supply?

Answer

The inferior wall and inferoseptum (segments 3, 4, 9, 10, 15) in a right-dominant system.

#290 · Interpretation of Nuclear Cardiology Imaging Tests

How is coronary dominance defined?

Answer

By the artery that gives rise to the posterior descending artery. About 85 percent of people are right dominant.

#291 · Interpretation of Nuclear Cardiology Imaging Tests

A patient is left dominant. Which artery most likely causes an inferior wall defect?

Answer

The left circumflex, because it supplies the posterior descending territory in left dominance.

#292 · Interpretation of Nuclear Cardiology Imaging Tests

Why can a vessel not be named with certainty from a perfusion defect alone?

Answer

Territories overlap and vary: the apex, basal inferolateral wall and distal LAD wrap-around differ between patients. The defect shows myocardium at risk.

#293 · Interpretation of Nuclear Cardiology Imaging Tests

What causes a small basal septal defect that is normal?

Answer

The membranous septum is thin fibrous tissue and has fewer myocytes, so counts are lower there.

#294 · Interpretation of Nuclear Cardiology Imaging Tests

What is apical thinning on perfusion imaging?

Answer

A normal fall-off in counts at the true apex, present at stress and rest, with normal thickening on gating.

#295 · Interpretation of Nuclear Cardiology Imaging Tests

A fixed mild inferior defect with normal thickening in an obese man most likely represents what?

Answer

Diaphragmatic attenuation. Normal wall thickening on gating favors artifact over scar.

#296 · Interpretation of Nuclear Cardiology Imaging Tests

Which walls are typically affected by breast attenuation?

Answer

The anterior, anterolateral and anteroseptal walls, depending on breast position, which may differ between stress and rest.

#297 · Interpretation of Nuclear Cardiology Imaging Tests

Define a reversible perfusion defect.

Answer

Reduced stress uptake that improves or normalizes at rest, indicating stress-induced ischemia.

#298 · Interpretation of Nuclear Cardiology Imaging Tests

What does a fixed defect usually mean, and what else can it be?

Answer

Usually scar. It can also be severe hibernation, attenuation or a technical issue, so check gating, attenuation correction or viability.

#299 · Interpretation of Nuclear Cardiology Imaging Tests

What is reverse redistribution?

Answer

A rest image worse than the stress image, seen with a partly reperfused infarct, after thrombolysis or intervention, or as a technical artifact.

#300 · Interpretation of Nuclear Cardiology Imaging Tests

What is balanced ischemia?

Answer

Severe three-vessel or left main disease reducing flow in all territories equally, so relative perfusion looks falsely normal.

#301 · Interpretation of Nuclear Cardiology Imaging Tests

What indirect clues suggest balanced ischemia on SPECT?

Answer

TID, a fall in EF with stress, increased lung uptake, symptoms, marked ST depression and hypotension.

#302 · Interpretation of Nuclear Cardiology Imaging Tests

Which imaging approach best exposes balanced ischemia?

Answer

PET with absolute myocardial blood flow and flow reserve, because it does not depend on relative uptake.

#303 · Interpretation of Nuclear Cardiology Imaging Tests

What is the typical false-positive pattern in left bundle branch block?

Answer

A reversible septal or anteroseptal defect with normal LAD angiography, more likely with exercise than with vasodilator stress.

#304 · Interpretation of Nuclear Cardiology Imaging Tests

Which stress is preferred in LBBB or ventricular pacing, and why?

Answer

Vasodilator stress, because the septal artifact is rate-related and less frequent without a heart rate rise.

#305 · Interpretation of Nuclear Cardiology Imaging Tests

In LBBB, which finding favors true LAD disease rather than the LBBB artifact?

Answer

A defect extending to the anterior wall or apex beyond the septum.

#306 · Interpretation of Nuclear Cardiology Imaging Tests

How does attenuation correction help in reading an inferior defect?

Answer

A defect that normalizes with correction and has normal thickening favors attenuation. A persistent defect with abnormal motion favors scar.

#307 · Interpretation of Nuclear Cardiology Imaging Tests

Why should uncorrected images be read alongside corrected images?

Answer

They show true counts and typical attenuation patterns, and correction can add its own artifacts.

#308 · Interpretation of Nuclear Cardiology Imaging Tests

What is the most common cause of a false lateral defect on CT-corrected PET?

Answer

Misregistration between the CT map and emission data, placing lateral myocardium over lung on the map.

#309 · Interpretation of Nuclear Cardiology Imaging Tests

What is the first action when an AC PET study shows an unexpected lateral defect?

Answer

Check and, if needed, manually realign the CT attenuation map with the emission images before reporting.

#310 · Interpretation of Nuclear Cardiology Imaging Tests

Define transient ischemic dilation.

Answer

The ratio of stress to rest LV cavity volume, elevated by diffuse subendocardial ischemia, a marker of severe extensive disease.

#311 · Interpretation of Nuclear Cardiology Imaging Tests

What are the approximate TID thresholds, and how should they be applied?

Answer

About 1.2 for same-day Tc-99m and about 1.3 or higher with Tl-201 protocols. Use the laboratory and software threshold.

#312 · Interpretation of Nuclear Cardiology Imaging Tests

Does a normal relative perfusion pattern with a high TID exclude severe disease?

Answer

No. It should raise concern for balanced ischemia from multivessel or left main disease.

#313 · Interpretation of Nuclear Cardiology Imaging Tests

What is the lung-heart ratio and its abnormal value on Tl-201?

Answer

Lung to myocardial count ratio on anterior planar images. Above about 0.5 is abnormal, reflecting stress LV dysfunction and raised filling pressure.

#314 · Interpretation of Nuclear Cardiology Imaging Tests

What is the difference between hibernating and stunned myocardium?

Answer

Hibernating is chronically underperfused but viable. Stunned is dysfunctional after transient ischemia although flow is restored.

#315 · Interpretation of Nuclear Cardiology Imaging Tests

What uptake level marks viability for Tl-201 or sestamibi?

Answer

About 50 to 60 percent of peak uptake in the normal region.

#316 · Interpretation of Nuclear Cardiology Imaging Tests

Why is Tl-201 reinjection used?

Answer

A small second dose after redistribution reveals viability in segments that look fixed.

#317 · Interpretation of Nuclear Cardiology Imaging Tests

Why does sestamibi underestimate viability, and how can accuracy improve?

Answer

It does not redistribute, so low-flow viable segments look fixed. Giving nitrates before the rest injection improves detection.

#318 · Interpretation of Nuclear Cardiology Imaging Tests

Patient preparation for FDG viability imaging?

Answer

Controlled metabolic state with an oral glucose load or hyperinsulinemic-euglycemic clamp so myocardium uses glucose.

#319 · Interpretation of Nuclear Cardiology Imaging Tests

What does an FDG perfusion mismatch mean?

Answer

Reduced perfusion with preserved FDG uptake is hibernating myocardium likely to recover after revascularization.

#320 · Interpretation of Nuclear Cardiology Imaging Tests

What does a perfusion FDG match mean?

Answer

Reduced perfusion and reduced FDG uptake indicate scar with little chance of functional recovery.

#321 · Interpretation of Nuclear Cardiology Imaging Tests

What does high blood pool and little myocardial FDG uptake in a diabetic patient mean?

Answer

A technically inadequate study from poor glucose control, not scar.

#322 · Interpretation of Nuclear Cardiology Imaging Tests

What did the STICH viability substudy teach?

Answer

Viability testing did not identify patients who benefited differently from bypass surgery, so viability is one input, not an automatic indication.

#323 · Interpretation of Nuclear Cardiology Imaging Tests

What are the half lives of Rb-82, N-13 ammonia and F-18 flurpiridaz?

Answer

Rb-82 about 76 seconds, N-13 ammonia about 10 minutes, flurpiridaz about 110 minutes.

#324 · Interpretation of Nuclear Cardiology Imaging Tests

How is myocardial flow reserve calculated?

Answer

MFR = stress MBF divided by rest MBF. Rest 0.8 and stress 1.6 mL/min/g gives 2.0.

#325 · Interpretation of Nuclear Cardiology Imaging Tests

What MFR values are generally considered abnormal?

Answer

Below about 2.0 is reduced and below about 1.5 markedly reduced, with laboratory variation.

#326 · Interpretation of Nuclear Cardiology Imaging Tests

How can high resting blood flow distort MFR?

Answer

A high rate-pressure product raises rest MBF and lowers MFR falsely, so correct rest flow when your laboratory does so.

#327 · Interpretation of Nuclear Cardiology Imaging Tests

What does a globally reduced MFR with a normal relative map suggest?

Answer

Multivessel disease or microvascular dysfunction, with a worse prognosis.

#328 · Interpretation of Nuclear Cardiology Imaging Tests

How does hypertrophic cardiomyopathy affect perfusion images?

Answer

Asymmetric septal hypertrophy gives high septal counts and a relatively low lateral wall, causing false defects. A small cavity inflates gated EF.

#329 · Interpretation of Nuclear Cardiology Imaging Tests

How does a dilated cardiomyopathy complicate interpretation?

Answer

The thin dilated ventricle gives uneven or patchy counts that resemble scar. Territorial defects favor ischemic disease.

#330 · Interpretation of Nuclear Cardiology Imaging Tests

What four properties describe every perfusion defect in a report?

Answer

Location, extent, severity and reversibility.

#331 · Interpretation of Nuclear Cardiology Imaging Tests

How is LVEF calculated from volumes?

Answer

LVEF = (EDV minus ESV) divided by EDV, times 100. If EDV is 140 mL and ESV is 70 mL, stroke volume is 70 mL and EF is 50 percent.

#332 · Interpretation of Nuclear Cardiology Imaging Tests

EDV 95 mL and ESV 35 mL. What is the EF?

Answer

Stroke volume is 60 mL and EF is 60 divided by 95, about 63 percent.

#333 · Interpretation of Nuclear Cardiology Imaging Tests

Which volume is the stronger prognostic predictor, EDV or ESV?

Answer

ESV, because it reflects systolic failure directly. A larger ESV and a lower EF both predict worse outcome.

#334 · Interpretation of Nuclear Cardiology Imaging Tests

How many frames per R-R cycle are typical for gated SPECT, and what does 16 frames improve?

Answer

Eight is the long-standing default. Sixteen frames sample the curve better, especially in small hearts and at fast rates, because 8 frames blur end-systole and underestimate the smallest volume.

#335 · Interpretation of Nuclear Cardiology Imaging Tests

What does the beat acceptance window do?

Answer

It accepts only beats whose R-R length falls within a set range, often 20 percent around the average, and rejects the rest.

#336 · Interpretation of Nuclear Cardiology Imaging Tests

Why should serial gated studies use the same software and frame number?

Answer

Programs and frame counts give different EF and volume values, so a change may reflect the method, not the patient.

#337 · Interpretation of Nuclear Cardiology Imaging Tests

Which SPECT tracers give the best gated function, and why is thallium poor for it?

Answer

Tc-99m sestamibi and tetrofosmin, because of high counts. Thallium-201 has low counts and low-energy photons, so gating is unreliable.

#338 · Interpretation of Nuclear Cardiology Imaging Tests

What advantage does gated PET have for stunning?

Answer

PET acquires during stress, so it can show LVEF at peak stress. A stress EF that fails to rise by at least 5 points, or falls, suggests severe multivessel disease.

#339 · Interpretation of Nuclear Cardiology Imaging Tests

What is the normal LVEF response from rest to stress on PET?

Answer

A rise of at least 5 points. A fall points to severe, usually multivessel, disease.

#340 · Interpretation of Nuclear Cardiology Imaging Tests

What is the difference between wall motion and wall thickening?

Answer

Motion is inward movement of the endocardial border in systole. Thickening is the increase in wall thickness, seen as a brightening of counts through partial volume effect.

#341 · Interpretation of Nuclear Cardiology Imaging Tests

State the 5-point wall motion scale.

Answer

0 normal, 1 mild hypokinesis, 2 moderate hypokinesis, 3 severe hypokinesis, 4 akinesis, 5 dyskinesis.

#342 · Interpretation of Nuclear Cardiology Imaging Tests

What does dyskinesis mean?

Answer

Paradoxical outward movement of a wall segment during systole, as in an aneurysm.

#343 · Interpretation of Nuclear Cardiology Imaging Tests

A fixed perfusion defect with normal motion and normal thickening is most likely what?

Answer

Attenuation artifact (breast or diaphragm), not transmural scar.

#344 · Interpretation of Nuclear Cardiology Imaging Tests

A fixed defect with akinesis and no thickening suggests what?

Answer

Infarct, most likely transmural, with thin nonthickening wall.

#345 · Interpretation of Nuclear Cardiology Imaging Tests

What is expected at rest in a reversible stress defect without prior infarct?

Answer

Normal rest wall motion, because the myocardium was not injured.

#346 · Interpretation of Nuclear Cardiology Imaging Tests

Why does LBBB or RV pacing not prove scar when septal motion is abnormal?

Answer

They cause abnormal septal motion with preserved septal thickening, so the segment is not scar.

#347 · Interpretation of Nuclear Cardiology Imaging Tests

Which is the better guide to viable muscle, motion or thickening?

Answer

Thickening. Motion can be paradoxical after bypass surgery, with LBBB, or with pacing while thickening stays normal.

#348 · Interpretation of Nuclear Cardiology Imaging Tests

Define post-stress stunning.

Answer

Post-stress LVEF or regional function worse than at rest despite viable myocardium, from delayed recovery after severe ischemia.

#349 · Interpretation of Nuclear Cardiology Imaging Tests

What EF change from rest to post-stress suggests severe extensive ischemia?

Answer

A fall of 5 points or more (some laboratories use 5 to 10), suggesting left main or multivessel disease.

#350 · Interpretation of Nuclear Cardiology Imaging Tests

Why is stunning more common after exercise than after vasodilator stress?

Answer

Exercise causes more true ischemia, and vasodilator stress causes less, so stunning when seen after vasodilator is a stronger signal.

#351 · Interpretation of Nuclear Cardiology Imaging Tests

A normal-looking perfusion study shows a post-stress EF drop. What do you suspect?

Answer

Balanced multivessel ischemia, because uptake falls evenly and relative perfusion looks normal.

#352 · Interpretation of Nuclear Cardiology Imaging Tests

Why is EF overestimated in a small heart?

Answer

The cavity is near camera resolution, so end-systolic volume is underestimated and EF is overestimated, sometimes by 10 points or more.

#353 · Interpretation of Nuclear Cardiology Imaging Tests

Name two settings that produce a small heart with an overestimated EF.

Answer

Hypertrophic cardiomyopathy or marked concentric LV hypertrophy, and small female hearts with EDV below about 70 mL.

#354 · Interpretation of Nuclear Cardiology Imaging Tests

How should a high EF in a small heart be reported?

Answer

With a caveat about overestimation, and recommend echocardiography or MRI if the EF matters.

#355 · Interpretation of Nuclear Cardiology Imaging Tests

What effect do low counts have on gated EF?

Answer

Edge detection fails and end-systolic volume is hard to trace, so EF becomes unreliable.

#356 · Interpretation of Nuclear Cardiology Imaging Tests

Which patient factor gives a false low EF when a very large perfusion defect is present?

Answer

Software edges may run through a zero-count region, so contour error can lower the EF.

#357 · Interpretation of Nuclear Cardiology Imaging Tests

What does atrial fibrillation do to gated SPECT?

Answer

Variable R-R lengths blur end-systole and cause many rejected beats, lowering counts and EF accuracy.

#358 · Interpretation of Nuclear Cardiology Imaging Tests

How do you judge whether gating was valid?

Answer

Check rejected beats and the beat-length histogram, a smooth time-volume curve with a minimum at about 30 to 40 percent of the R-R, and contours on the endocardium.

#359 · Interpretation of Nuclear Cardiology Imaging Tests

What should be done when gating is poor?

Answer

State it in the report and qualify or omit the EF. Do not call wall motion normal.

#360 · Interpretation of Nuclear Cardiology Imaging Tests

Why is RV EF unreliable on standard gated perfusion SPECT?

Answer

The RV is thin and crescent shaped, so it cannot be measured reliably. Use first-pass, MUGA, echocardiography or MRI.

#361 · Interpretation of Nuclear Cardiology Imaging Tests

What is the approximate lower limit of normal RVEF by radionuclide methods?

Answer

About 40 percent.

#362 · Interpretation of Nuclear Cardiology Imaging Tests

What does MUGA image, and why does that matter for EF?

Answer

The blood pool. LV counts are proportional to volume, so EF comes from counts without geometric assumptions.

#363 · Interpretation of Nuclear Cardiology Imaging Tests

Which labeling method gives the highest red cell labeling efficiency?

Answer

In vitro kit labeling such as UltraTag, above 95 percent. Modified in vivo is about 85 percent and in vivo about 75 to 80 percent.

#364 · Interpretation of Nuclear Cardiology Imaging Tests

What does a bright stomach and thyroid on MUGA indicate?

Answer

Poor red cell labeling with free pertechnetate.

#365 · Interpretation of Nuclear Cardiology Imaging Tests

What is a typical Tc-99m dose for MUGA?

Answer

About 20 to 30 mCi (740 to 1,110 MBq).

#366 · Interpretation of Nuclear Cardiology Imaging Tests

Which MUGA view is used to calculate LVEF?

Answer

The best septal LAO view, about 30 to 45 degrees, with the septum vertical and LV and RV separated, often with caudal tilt.

#367 · Interpretation of Nuclear Cardiology Imaging Tests

Name the three standard MUGA views.

Answer

Anterior, best septal LAO, and left lateral or left posterior oblique.

#368 · Interpretation of Nuclear Cardiology Imaging Tests

Write the MUGA EF formula.

Answer

EF = (ED net counts minus ES net counts) divided by ED net counts, with background subtracted.

#369 · Interpretation of Nuclear Cardiology Imaging Tests

ED counts 120,000, ES counts 60,000, background 20,000. EF with background correction?

Answer

Net ED 100,000, net ES 40,000, EF is 60,000 divided by 100,000 = 60 percent. Without background it would read 50 percent.

#370 · Interpretation of Nuclear Cardiology Imaging Tests

What happens to MUGA EF if background is set too high?

Answer

Over-subtraction lowers net end-systolic counts proportionally more, so EF is falsely raised.

#371 · Interpretation of Nuclear Cardiology Imaging Tests

Where should the MUGA background region not be placed?

Answer

Over the spleen, left atrium or aorta, which gives falsely high background and a falsely high EF.

#372 · Interpretation of Nuclear Cardiology Imaging Tests

What does phase analysis show?

Answer

The timing (phase) and amplitude of contraction from the first Fourier harmonic of each pixel's time-activity curve, with a narrow histogram when contraction is synchronous.

#373 · Interpretation of Nuclear Cardiology Imaging Tests

What does LBBB do to the phase histogram?

Answer

Delays the LV, especially the lateral wall, and widens the histogram.

#374 · Interpretation of Nuclear Cardiology Imaging Tests

Does phase analysis select patients for cardiac resynchronization therapy?

Answer

No. Guidelines use EF, QRS duration and QRS morphology, not phase analysis.

#375 · Interpretation of Nuclear Cardiology Imaging Tests

What EF change defines cardiotoxicity during chemotherapy?

Answer

A fall of 10 points or more to below the lower limit of normal, about 50 to 53 percent.

#376 · Interpretation of Nuclear Cardiology Imaging Tests

Which test is standard for suspected cardiac sarcoidosis?

Answer

F-18 FDG PET read together with a resting perfusion study.

#377 · Interpretation of Nuclear Cardiology Imaging Tests

How is normal myocardial FDG uptake suppressed before a sarcoid scan?

Answer

A high-fat, very low-carbohydrate diet for at least 24 hours followed by a fast of at least 12 hours, with or without heparin before FDG.

#378 · Interpretation of Nuclear Cardiology Imaging Tests

What does normal perfusion with focal FDG uptake suggest in sarcoid?

Answer

Early active inflammation before scar or microvascular compression has developed.

#379 · Interpretation of Nuclear Cardiology Imaging Tests

What does a perfusion defect with FDG uptake in the same area mean?

Answer

A mismatch pattern: active inflammation, often with compression of small vessels.

#380 · Interpretation of Nuclear Cardiology Imaging Tests

What does a perfusion defect without FDG uptake mean in sarcoid?

Answer

Scar, or burned-out disease.

#381 · Interpretation of Nuclear Cardiology Imaging Tests

A sarcoid FDG scan shows diffuse uptake in the whole left ventricle. What is the usual cause?

Answer

Failed suppression of normal glucose use from poor diet or fast, or diabetes. It should not be called sarcoid.

#382 · Interpretation of Nuclear Cardiology Imaging Tests

What does a falling extent of FDG uptake on follow-up show in sarcoid?

Answer

Response to immunosuppressive treatment.

#383 · Interpretation of Nuclear Cardiology Imaging Tests

Which tracer is used for ATTR cardiac amyloid imaging?

Answer

Tc-99m pyrophosphate (PYP), 10 to 20 mCi (370 to 740 MBq).

#384 · Interpretation of Nuclear Cardiology Imaging Tests

Why must SPECT be done for a PYP study?

Answer

Planar images cannot separate myocardial uptake from blood pool activity. SPECT or SPECT/CT does.

#385 · Interpretation of Nuclear Cardiology Imaging Tests

How are PYP visual grades defined?

Answer

Grade 0 none, 1 less than rib, 2 equal to rib, 3 greater than rib with mild or absent rib uptake. Grade 2 or 3 is positive.

#386 · Interpretation of Nuclear Cardiology Imaging Tests

What 1-hour heart-to-contralateral ratio is abnormal on PYP?

Answer

1.5 or greater.

#387 · Interpretation of Nuclear Cardiology Imaging Tests

What 3-hour heart-to-contralateral ratio is abnormal on PYP?

Answer

1.3 or greater.

#388 · Interpretation of Nuclear Cardiology Imaging Tests

A PYP heart region has 4,200 counts and the mirror region has 3,000. What is the ratio?

Answer

4,200 divided by 3,000 = 1.4, which is abnormal on a 3-hour image and below the 1-hour cutoff.

#389 · Interpretation of Nuclear Cardiology Imaging Tests

When can ATTR amyloid be diagnosed without biopsy?

Answer

With SPECT-confirmed grade 2 or 3 PYP uptake and no monoclonal protein on serum free light chains and serum and urine immunofixation.

#390 · Interpretation of Nuclear Cardiology Imaging Tests

PYP is positive and a monoclonal protein is present. What is the next step?

Answer

Tissue biopsy to exclude AL amyloid, which needs different and urgent treatment.

#391 · Interpretation of Nuclear Cardiology Imaging Tests

Planar PYP looks positive, but SPECT shows only ventricular cavity activity. How do you read it?

Answer

Negative for myocardial uptake. The planar finding was blood pool.

#392 · Interpretation of Nuclear Cardiology Imaging Tests

Name the main uses of FDG PET/CT in cardiac infection.

Answer

Suspected cardiac device infection and prosthetic valve endocarditis when echo and cultures are not enough.

#393 · Interpretation of Nuclear Cardiology Imaging Tests

Why is early postoperative FDG uptake around a valve hard to interpret?

Answer

Postoperative inflammation causes uptake, so timing and pattern matter. Focal intense uptake at the ring is more specific.

#394 · Interpretation of Nuclear Cardiology Imaging Tests

Which alternative to FDG is used for infection imaging?

Answer

Autologous leukocytes labeled with Tc-99m HMPAO, acquired with SPECT/CT.

#395 · Interpretation of Nuclear Cardiology Imaging Tests

What is I-123 MIBG?

Answer

A norepinephrine analog taken up and stored by sympathetic nerve terminals, used to assess cardiac sympathetic innervation.

#396 · Interpretation of Nuclear Cardiology Imaging Tests

What is the usual I-123 MIBG dose?

Answer

About 10 mCi (370 MBq) intravenously.

#397 · Interpretation of Nuclear Cardiology Imaging Tests

Why is potassium iodide given before MIBG?

Answer

To block thyroid uptake of free radioiodine.

#398 · Interpretation of Nuclear Cardiology Imaging Tests

When are MIBG planar images acquired?

Answer

Early at about 15 minutes and late at about 4 hours.

#399 · Interpretation of Nuclear Cardiology Imaging Tests

What is a normal late heart-to-mediastinum ratio on MIBG?

Answer

About 2.2 or greater.

#400 · Interpretation of Nuclear Cardiology Imaging Tests

Which late H/M ratio is abnormal on MIBG?

Answer

Below 1.6, which signals higher risk in heart failure.

#401 · Interpretation of Nuclear Cardiology Imaging Tests

Early MIBG heart count is 100 and mediastinum is 50. Late heart is 70 and mediastinum is 50. What are the H/M ratios?

Answer

Early 100 divided by 50 = 2.0. Late 70 divided by 50 = 1.4, which is abnormal.

#402 · Interpretation of Nuclear Cardiology Imaging Tests

What does increased MIBG washout mean?

Answer

Increased sympathetic drive and a worse prognosis.

#403 · Interpretation of Nuclear Cardiology Imaging Tests

Which drugs interfere with MIBG uptake?

Answer

Labetalol, tricyclic antidepressants and some sympathomimetics, so review the medication list.

#404 · Interpretation of Nuclear Cardiology Imaging Tests

What are the Agatston coronary calcium categories?

Answer

0 none, 1 to 99 mild, 100 to 399 moderate, 400 or greater severe.

#405 · Interpretation of Nuclear Cardiology Imaging Tests

Can the attenuation correction CT be used for a calcium score?

Answer

No. It is ungated and free-breathing. A separate gated noncontrast CT is needed.

#406 · Interpretation of Nuclear Cardiology Imaging Tests

A patient has normal perfusion but CAC of 600. What should the report say?

Answer

Perfusion is normal, plaque burden is high, and intensive preventive therapy is supported. Normal perfusion does not mean normal arteries.

#407 · Interpretation of Nuclear Cardiology Imaging Tests

What does MPI measure that CAC does not?

Answer

Flow-limiting disease. CAC measures anatomic plaque burden.

#408 · Interpretation of Nuclear Cardiology Imaging Tests

What are the summed stress score ranges?

Answer

Below 4 normal, 4 to 8 mild, 9 to 13 moderate, above 13 severe.

#409 · Interpretation of Nuclear Cardiology Imaging Tests

How is the summed difference score calculated?

Answer

SSS minus SRS. It reflects ischemia.

#410 · Interpretation of Nuclear Cardiology Imaging Tests

What does the summed rest score reflect?

Answer

Scar or infarcted myocardium.

#411 · Interpretation of Nuclear Cardiology Imaging Tests

SSS is 15 and SRS is 7. What are SDS and percent ischemic myocardium?

Answer

SDS = 8. Percent = 8 divided by 68 = about 12 percent.

#412 · Interpretation of Nuclear Cardiology Imaging Tests

What annual event rate follows a normal perfusion study?

Answer

Below 1 percent per year for cardiac death or infarction.

#413 · Interpretation of Nuclear Cardiology Imaging Tests

Which patients have a shorter warranty after a normal study?

Answer

Diabetic patients, those with prior revascularization and those with high calcium score.

#414 · Interpretation of Nuclear Cardiology Imaging Tests

What ischemic burden suggested revascularization benefit in observational nuclear data?

Answer

About 10 percent or more of the myocardium.

#415 · Interpretation of Nuclear Cardiology Imaging Tests

What did the ISCHEMIA trial find?

Answer

In stable patients with moderate to severe ischemia, initial invasive strategy did not reduce death or MI compared with optimal medical therapy.

#416 · Interpretation of Nuclear Cardiology Imaging Tests

Calculate the Duke score for 9 minutes of exercise, 2 mm of ST depression and an angina index of 1.

Answer

9 - (5 x 2) - (4 x 1) = -5, which falls in the intermediate risk group (-10 to +4).

#417 · Interpretation of Nuclear Cardiology Imaging Tests

A patient has a Duke treadmill score of -12. Which risk group is that?

Answer

High risk, because the group is defined as -11 or lower; this usually points toward invasive evaluation rather than reassurance.

#418 · Interpretation of Nuclear Cardiology Imaging Tests

Exercise 10 minutes, ST 2 mm, nonlimiting angina (index 1). What is the Duke score?

Answer

10 minus 10 minus 4 = minus 4, intermediate risk.

#419 · Interpretation of Nuclear Cardiology Imaging Tests

Which heart rate recovery is abnormal?

Answer

A fall of 12 beats per minute or less in the first minute after exercise with an upright cool-down.

#420 · Interpretation of Nuclear Cardiology Imaging Tests

What does a high TID ratio suggest?

Answer

Severe, extensive ischemia, but cutoffs vary from about 1.2 to 1.4. Check technique when the perfusion is otherwise clean.

#421 · Interpretation of Nuclear Cardiology Imaging Tests

An exercise study in a patient with LBBB shows an isolated septal defect. What is the best approach?

Answer

Use a vasodilator for stress, and do not call the septal defect LAD ischemia without considering the LBBB artifact.

#422 · Interpretation of Nuclear Cardiology Imaging Tests

When is preoperative stress imaging reasonable?

Answer

In elevated risk surgery with poor or unknown functional capacity when the result would change management.

#423 · Interpretation of Nuclear Cardiology Imaging Tests

Which vessel territories do the LAD and RCA supply on the 17-segment model?

Answer

LAD: anterior, anteroseptal and apex. RCA: inferior and inferoseptal. The circumflex supplies the lateral wall.

#424 · Interpretation of Nuclear Cardiology Imaging Tests

What must you do for a critical result such as severe extensive ischemia?

Answer

Contact the ordering clinician directly, state the finding and recommendation, and document who and when.

#425 · Interpretation of Nuclear Cardiology Imaging Tests

What does a complete nuclear cardiology report include beyond the perfusion findings?

Answer

Indication, stress type, drugs and doses, symptoms, hemodynamics and ECG, radiopharmaceutical and dose, image quality, LVEF, TID and a final impression that answers the referring question.

#426 · Radiation Safety and Management of Radiopharmaceuticals

How many MBq are in 1 mCi?

Answer

1 mCi equals 37 MBq. One Ci equals 37 GBq.

#427 · Radiation Safety and Management of Radiopharmaceuticals

How many rad equal 1 Gy, and how many rem equal 1 Sv?

Answer

1 Gy equals 100 rad and 1 Sv equals 100 rem. So 1 mSv equals 100 mrem.

#428 · Radiation Safety and Management of Radiopharmaceuticals

Which dose quantity lets you compare a Tl-201 study with a chest CT?

Answer

Effective dose, the sum of organ equivalent doses times tissue weighting factors.

#429 · Radiation Safety and Management of Radiopharmaceuticals

What is the radiation weighting factor for photons and positrons versus alpha particles?

Answer

Photons and positrons are 1, so Gy equals Sv numerically. Alpha particles are 20.

#430 · Radiation Safety and Management of Radiopharmaceuticals

Give the physical half lives of Tc-99m, Tl-201 and F-18.

Answer

Tc-99m is 6.0 hours, Tl-201 is 73 hours and F-18 is 110 minutes.

#431 · Radiation Safety and Management of Radiopharmaceuticals

Give the half lives of Rb-82 and N-13.

Answer

Rb-82 is about 76 seconds and N-13 is about 10 minutes.

#432 · Radiation Safety and Management of Radiopharmaceuticals

How do you compute remaining activity after time t?

Answer

A = A0 x 0.5 raised to (t divided by half life). For Tc-99m, 30 mCi after 6 hours is about 15 mCi.

#433 · Radiation Safety and Management of Radiopharmaceuticals

You need 25 mCi of Tc-99m at 11:00 and the vial was assayed at 07:00. What was the 07:00 activity needed?

Answer

The decay factor over 4 hours is 0.630, so 25 divided by 0.630 is about 39.7 mCi.

#434 · Radiation Safety and Management of Radiopharmaceuticals

How do you combine physical and biological half lives?

Answer

1/Teff = 1/Tphysical + 1/Tbiological. The effective half life is shorter than either one.

#435 · Radiation Safety and Management of Radiopharmaceuticals

What is a half value layer?

Answer

The thickness of material that reduces beam intensity by half. For Tc-99m 140 keV photons in lead it is about 0.27 mm.

#436 · Radiation Safety and Management of Radiopharmaceuticals

What fraction of Tc-99m photons passes through 1 mm of lead?

Answer

1 mm is about 3.7 HVLs, so transmission is 0.5^3.7, about 8 percent.

#437 · Radiation Safety and Management of Radiopharmaceuticals

Why is shielding PET harder than shielding Tc-99m SPECT?

Answer

The 511 keV annihilation photons have a lead HVL near 4 mm, so tungsten shields and distance are used.

#438 · Radiation Safety and Management of Radiopharmaceuticals

A source reads 5 mR/h at 0.5 m. What is the reading at 2 m?

Answer

The inverse square law gives 5 x (0.5/2)^2, about 0.31 mR/h.

#439 · Radiation Safety and Management of Radiopharmaceuticals

What happens to exposure when you double your distance from a point source?

Answer

It falls to one quarter, because exposure varies as 1 divided by distance squared.

#440 · Radiation Safety and Management of Radiopharmaceuticals

Define stochastic effects of radiation.

Answer

Cancer and heritable effects whose probability rises with dose, whose severity does not, and which are assumed to have no threshold.

#441 · Radiation Safety and Management of Radiopharmaceuticals

Define deterministic effects and give an example threshold.

Answer

Tissue reactions with a threshold whose severity rises with dose, such as skin erythema above about 2 Gy. They do not occur at diagnostic imaging doses.

#442 · Radiation Safety and Management of Radiopharmaceuticals

What is the approximate effective dose of a one day rest and stress Tc-99m sestamibi study?

Answer

About 10 to 12 mSv.

#443 · Radiation Safety and Management of Radiopharmaceuticals

What is the approximate effective dose of an N-13 ammonia rest and stress PET study?

Answer

About 2 mSv.

#444 · Radiation Safety and Management of Radiopharmaceuticals

About what effective dose does a Rb-82 rest and stress study give?

Answer

About 2 to 4 mSv total.

#445 · Radiation Safety and Management of Radiopharmaceuticals

Roughly how does Tl-201 stress and redistribution dose compare with Tc-99m agents?

Answer

It is higher, roughly in the high teens to 20s of mSv at 3 mCi, so it is rarely the dose lowering choice.

#446 · Radiation Safety and Management of Radiopharmaceuticals

Calculate the effective dose for 370 MBq rest and 1,110 MBq stress Tc-99m using 0.0090 and 0.0079 mSv per MBq.

Answer

Rest is 3.3 mSv and stress is 8.8 mSv, totaling about 12.1 mSv.

#447 · Radiation Safety and Management of Radiopharmaceuticals

What is the NRC annual total effective dose limit for an adult radiation worker?

Answer

5 rem (50 mSv) per year.

#448 · Radiation Safety and Management of Radiopharmaceuticals

What are the NRC annual limits for the lens of the eye and for skin or extremities?

Answer

Lens is 15 rem (150 mSv) and skin or extremities is 50 rem (500 mSv).

#449 · Radiation Safety and Management of Radiopharmaceuticals

What are the NRC dose limits for members of the public?

Answer

0.1 rem (1 mSv) per year, and no more than 2 mrem (0.02 mSv) in any one hour in an unrestricted area.

#450 · Radiation Safety and Management of Radiopharmaceuticals

What dose limit applies to the embryo or fetus of a declared pregnant worker?

Answer

0.5 rem (5 mSv) for the entire pregnancy, spread evenly, about 0.05 rem (0.5 mSv) or less in a month.

#451 · Radiation Safety and Management of Radiopharmaceuticals

When does a worker become a declared pregnant worker?

Answer

Only after she tells her employer in writing and gives the estimated date of conception.

#452 · Radiation Safety and Management of Radiopharmaceuticals

Is there a dose limit for a patient receiving a medically justified nuclear study?

Answer

No. The study is justified by the physician and ALARA still applies.

#453 · Radiation Safety and Management of Radiopharmaceuticals

What are the three ALARA tools?

Answer

Time, distance and shielding.

#454 · Radiation Safety and Management of Radiopharmaceuticals

What do you do first when a radioactive liquid spill occurs?

Answer

Warn others, confine the area, notify the radiation safety officer, then clean from the outside edge inward and survey.

#455 · Radiation Safety and Management of Radiopharmaceuticals

How should contaminated skin be decontaminated?

Answer

Wash with mild soap and lukewarm water without abrasive scrubbing, then survey.

#456 · Radiation Safety and Management of Radiopharmaceuticals

What is the Mo-99 breakthrough limit for Tc-99m eluate?

Answer

0.15 microcurie of Mo-99 per millicurie of Tc-99m (0.15 kBq per MBq) at administration.

#457 · Radiation Safety and Management of Radiopharmaceuticals

What is the aluminum breakthrough limit in generator eluate?

Answer

10 micrograms per mL.

#458 · Radiation Safety and Management of Radiopharmaceuticals

How often are dose calibrator constancy, linearity and accuracy tested?

Answer

Constancy daily, linearity quarterly and accuracy annually (also after repair).

#459 · Radiation Safety and Management of Radiopharmaceuticals

When can short lived waste be discarded as ordinary waste?

Answer

After decay in storage for at least 10 half lives with a survey to background; for Tc-99m that is about 60 hours.

#460 · Radiation Safety and Management of Radiopharmaceuticals

What do you tell a woman about breast feeding after Tl-201?

Answer

Interrupt breast feeding for about 2 weeks and check current NRC Regulatory Guide 8.39 and facility policy.

#461 · Radiation Safety and Management of Radiopharmaceuticals

What written advice is given to a patient after imaging?

Answer

A card with the radiopharmaceutical, date and activity, since radiation monitors at borders may detect it for days to weeks.

#462 · Radiation Safety and Management of Radiopharmaceuticals

What contact advice is given to patients after a nuclear scan?

Answer

Avoid prolonged close contact with infants under 1 year and pregnant women for the interval set by policy, and flush the toilet twice.

#463 · Radiation Safety and Management of Radiopharmaceuticals

A diagnostic wrong patient dose error occurs. What do you do?

Answer

Assess the patient, document, notify the supervising physician, the RSO and referring clinician, and inform the patient; most are not NRC medical events.

#464 · Radiation Safety and Management of Radiopharmaceuticals

When must an NRC reportable medical event be reported by phone and in writing?

Answer

By phone by the next calendar day and in writing within 15 days, with the patient informed within 24 hours.

#465 · Radiation Safety and Management of Radiopharmaceuticals

Who is responsible for the overall radiation safety program under the license, and who for clinical use?

Answer

The RSO oversees the program, and the authorized user (physician) is responsible for clinical use and supervision of technologists.

#466 · Radiation Safety and Management of Radiopharmaceuticals

Within what time must a received radioactive package be surveyed?

Answer

Within 3 hours of receipt during normal working hours, or within 3 hours of the start of the next working day if it arrives after hours (10 CFR 20.1906).

#467 · Radiation Safety and Management of Radiopharmaceuticals

What package dose rates trigger action on receipt?

Answer

200 mrem per hour (2 mSv per hour) at the surface or 10 mrem per hour (0.1 mSv per hour) at 1 meter. Notify the radiation safety officer, and the NRC when the limits are exceeded.

#468 · Radiation Safety and Management of Radiopharmaceuticals

How often are meter surveys and wipe tests done in the hot lab?

Answer

A meter survey at the end of each day of use and weekly wipe tests where unsealed material is prepared, given or stored (10 CFR 35.70).

#469 · Radiation Safety and Management of Radiopharmaceuticals

How long is short lived waste held for decay in storage?

Answer

At least 10 half-lives, then a surface survey must show no reading distinguishable from background and the result is logged. For Tc-99m this is about 60 hours.

#470 · Radiation Safety and Management of Radiopharmaceuticals

What is the half-life of Mo-99 and what fraction of its decays feed Tc-99m?

Answer

About 66 hours, and about 88 percent of decays go through the Tc-99m branch.

#471 · Radiation Safety and Management of Radiopharmaceuticals

When does Tc-99m activity peak after a generator elution?

Answer

At about 23 hours (transient equilibrium). Waiting longer gains no more Tc-99m, so daily elution gives the most activity.

#472 · Radiation Safety and Management of Radiopharmaceuticals

What is the NRC limit for Mo-99 breakthrough in Tc-99m eluate?

Answer

0.15 microcuries of Mo-99 per millicurie of Tc-99m (0.15 kBq per MBq) at the time of administration.

#473 · Radiation Safety and Management of Radiopharmaceuticals

How is Mo-99 breakthrough measured in the dose calibrator?

Answer

Put the vial in a lead shield about 6 mm thick. It stops the 140 keV Tc-99m photon and passes the 740 and 778 keV Mo-99 photons.

#474 · Radiation Safety and Management of Radiopharmaceuticals

An eluate has 0.10 microcuries of Mo-99 per millicurie at elution. What is the ratio 6 hours later?

Answer

About 0.188 microcuries per millicurie: Tc-99m falls to 0.50 while Mo-99 falls only to 0.939, so 0.10 x 0.939 / 0.50. It fails the 0.15 limit.

#475 · Radiation Safety and Management of Radiopharmaceuticals

What aluminum ion limit applies to generator eluate?

Answer

10 micrograms per milliliter (USP limit). Excess aluminum interferes with labeling and clumps sulfur colloid.

#476 · Radiation Safety and Management of Radiopharmaceuticals

What radiochemical purity is required for sestamibi and tetrofosmin?

Answer

90 percent or more, checked by thin layer chromatography with the kit's own method.

#477 · Radiation Safety and Management of Radiopharmaceuticals

What are the radiochemical impurities of Tc-99m kits?

Answer

Free pertechnetate and reduced hydrolyzed technetium, an insoluble colloid trapped in the liver.

#478 · Radiation Safety and Management of Radiopharmaceuticals

A kit fails the radiochemical purity test. What do you do?

Answer

Discard it and prepare a new kit. Do not inject a different amount of the failed preparation.

#479 · Radiation Safety and Management of Radiopharmaceuticals

What is the formula for decay correction?

Answer

A = A0 x 2 to the minus (t divided by half-life), with t and half-life in the same units.

#480 · Radiation Safety and Management of Radiopharmaceuticals

Give the half-lives of Tc-99m, Tl-201, F-18, N-13 and Rb-82.

Answer

Tc-99m 6.02 hours, Tl-201 73 hours, F-18 109.8 minutes, N-13 9.97 minutes, Rb-82 about 76 seconds.

#481 · Radiation Safety and Management of Radiopharmaceuticals

Eluate is 10 mCi per mL at 08:00. What volume gives 10 mCi at 11:00?

Answer

After 3 hours the factor is 0.708, so concentration is 7.08 mCi per mL and the volume is 1.41 mL.

#482 · Radiation Safety and Management of Radiopharmaceuticals

What activity remains from 10 mCi of F-18 FDG after 140 minutes?

Answer

140 / 109.8 = 1.275 half-lives, factor 0.4132, so 4.13 mCi remains.

#483 · Radiation Safety and Management of Radiopharmaceuticals

What fraction of Rb-82 activity remains after 7.5 minutes?

Answer

About six half-lives have passed, so 1/64, about 1.6 percent.

#484 · Radiation Safety and Management of Radiopharmaceuticals

How often is dose calibrator constancy tested and what is it?

Answer

Daily, with a long lived source such as Cs-137 or Co-57; the reading must agree with the expected value within 5 percent.

#485 · Radiation Safety and Management of Radiopharmaceuticals

What are the other dose calibrator tests and their frequency?

Answer

Linearity quarterly, accuracy annually, and geometry at installation or after repair (10 CFR 35.60).

#486 · Radiation Safety and Management of Radiopharmaceuticals

What parent and contaminants are tested in an Rb-82 generator?

Answer

Sr-82 and Sr-85, which are bone seekers; the generator parent is Sr-82 (half-life 25.5 days).

#487 · Radiation Safety and Management of Radiopharmaceuticals

What are the NRC limits for Rb-82 generator eluate?

Answer

0.02 microcuries of Sr-82 and 0.2 microcuries of Sr-85 per millicurie of Rb-82.

#488 · Radiation Safety and Management of Radiopharmaceuticals

What do you do when an Rb-82 eluate exceeds the strontium limit?

Answer

Stop using the generator, notify the manufacturer and the radiation safety officer, and follow label directions for patient follow up.

#489 · Radiation Safety and Management of Radiopharmaceuticals

Describe the in vivo RBC labeling method for MUGA.

Answer

Inject stannous pyrophosphate, wait 15 to 20 minutes, then inject 20 to 25 mCi of Tc-99m pertechnetate. Efficiency is about 75 to 80 percent.

#490 · Radiation Safety and Management of Radiopharmaceuticals

Describe the modified in vivo RBC labeling method.

Answer

Inject stannous pyrophosphate, wait about 20 minutes, draw 3 to 5 mL of blood into a syringe with pertechnetate, incubate about 10 minutes and reinject. Efficiency is about 85 to 90 percent.

#491 · Radiation Safety and Management of Radiopharmaceuticals

Which RBC labeling method gives the highest efficiency and why?

Answer

The in vitro kit method (over 95 percent); a hypochlorite step oxidizes extracellular tin so only the cell reduces the pertechnetate.

#492 · Radiation Safety and Management of Radiopharmaceuticals

A MUGA shows thyroid, salivary and stomach activity. What is the cause?

Answer

Free pertechnetate from poor red cell labeling, not a cardiac finding. Consider interfering drugs or a stannous problem.

#493 · Radiation Safety and Management of Radiopharmaceuticals

Name drugs that can reduce RBC labeling efficiency.

Answer

Heparin, doxorubicin, hydralazine, digoxin and iodinated contrast.

#494 · Radiation Safety and Management of Radiopharmaceuticals

Which administrations fall under the NRC medical event definition?

Answer

Those requiring a written directive that deliver a dose differing by more than 20 percent plus a dose threshold, or wrong drug, route or patient with that threshold (10 CFR 35.3045).

#495 · Radiation Safety and Management of Radiopharmaceuticals

What dose thresholds apply to an NRC medical event?

Answer

Effective dose over 0.05 Sv (5 rem) or dose to an organ or tissue over 0.5 Sv (50 rem).

#496 · Radiation Safety and Management of Radiopharmaceuticals

Is a wrong dose of Tc-99m sestamibi an NRC medical event?

Answer

Generally no, because diagnostic doses do not require a written directive. It is still a misadministration to document, review and report per state and institutional rules.

#497 · Radiation Safety and Management of Radiopharmaceuticals

Who is notified first after a wrong patient injection?

Answer

The supervising physician and the radiation safety officer, then the referring physician and the patient.

#498 · Radiation Safety and Management of Radiopharmaceuticals

What notice times apply to an NRC medical event?

Answer

Referring physician and patient within 24 hours, NRC by telephone by the next calendar day, and a written report within 15 days.

#499 · Radiation Safety and Management of Radiopharmaceuticals

When must a fetal or nursing child dose be reported to the NRC?

Answer

By the next calendar day when an embryo or fetus gets over 50 mSv (5 rem), or a nursing child gets an unintended dose over 5 mSv (0.5 rem) or the mother must stop nursing.

#500 · Radiation Safety and Management of Radiopharmaceuticals

What are the annual occupational and public dose limits?

Answer

Worker 50 mSv (5 rem), lens 150 mSv, skin and extremities 500 mSv, declared pregnant worker 5 mSv over the pregnancy, public 1 mSv (0.1 rem).